Safari – AI Powered Tour Guide Booking System with Voice and Retrieval Augmented Generation

25-26J-150

Project Final Thesis

B.Sc. (Hons) Information Technology in

Information Systems Engineering

Department of Computer Systems Engineering

Sri Lanka Institute of Information

Technology Sri Lanka

April 2026

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PRAVEAN P.R.I

(IT21191060)

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Name	Student ID	Signature
P.R.I. PRAVEAN	IT21191060	Inrajith

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

....

Signature of the supervisor:

(Ms. Suranjini Silva)

Date: 10/04/2026

....

Signature of the Co-supervisor:

(Dr. Anuradha Jayakody)

Date: 10/04/2026

Abstract

Tourism plays a vital role in the Sri Lankan economy, attracting millions of visitors each year who seek its cultural heritage, natural beauty, and diverse experiences. However, one of the major pain points faced by tourists is the difficulty in finding trustworthy, safe, and suitable tour guides. Existing online platforms such as TripAdvisor, Viator, and GetYourGuide provide only basic functionalities like listings, reviews, and ratings. These systems often fall short in three critical areas: verification, personalization, and safety. Tourists continue to face challenges such as unverified guides, misleading reviews, mismatched services, and premature exposure of personal details. This lack of reliability reduces trust in digital booking systems and can compromise the overall tourist experience, as well as the reputation of the destination.

To address these shortcomings, this research proposes the development of an AI Powered Tour Guide Booking System with Voice and Retrieval Augmented Generation (RAG). The platform is designed as a secure, intelligent, and user friendly web application that combines multi level verification, AI driven personalization, and privacy preserving workflows. Unlike traditional booking systems, our solution ensures that guides undergo rigorous verification processes using multiple documents, including government IDs, licenses, police checks, and optional social media validation. This provides tourists with confidence that they are interacting with legitimate, trusted guides.

The system also integrates advanced AI based matching and recommendation features, powered by RAG, to deliver personalized guide suggestions that go beyond static ratings. By analyzing tourist preferences, cultural contexts, past booking history, and even real time data such as weather or festivals, the platform ensures a truly tailored experience. In addition, a multilingual AI chat agent, supporting in English, enables tourists to make inquiries in their English language via voice or text, ensuring accessibility across diverse audiences. The system's scalability allows it to be expanded beyond Sri Lanka, positioning it as a global digital tourism innovation. By combining technological advancements with the core values of safety, trust, and personalization, the solution not only enhances individual travel experiences but also contributes to strengthening Sri Lanka’s reputation as a reliable and smart travel destination.

In summary, this project introduces a next generation tourism platform that bridges the gap between tourists’ expectations and the limitations of current systems. By integrating AI, RAG, and privacy focused workflows into a single solution, the AI Powered Tour Guide Booking System provides a balanced trust model where safety, personalization, and user satisfaction are at the forefront. This ensures that every journey is not only memorable but also secure, setting a new standard for the future of digital tourism.

Keywords—AI agents, AI-assisted tourism technology, booking platforms, tourism

List of Abbreviations

LLM	Large Language Model
DB	Database
SQL	Structured Query Language
AI	Artificial Intelligence
UI/UX	User Interface and User Experience
GDPR	General Data Protection Regulation
API	Application Programming Interface
IDE	Integrated Development Environment
NLP	Natural Language Processing
HTTP	Hyper Text Transfer Protocol
UAT	User Acceptance Testing
API	Application Programming Interface
GPU	Graphics Processing Unit
RAG	Retrieval Augmented Generation

Table1: List of Abbreviation

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1. Introduction

1.1 Background

The tourism industry plays a vital role in the economic development of many countries, particularly in emerging economies such as Sri Lanka. As a nation rich in natural beauty, cultural heritage, biodiversity, and historical significance, Sri Lanka has consistently attracted both international and domestic tourists. The sector significantly contributes to national income, foreign exchange earnings, employment generation, and regional development. According to existing literature, tourism is among the top contributors to Sri Lanka’s export economy, emphasizing its strategic importance in national growth and sustainability. Despite its economic significance, the tourism industry in Sri Lanka has experienced several disruptions over the years, including the Easter Sunday attacks in 2019 and the global COVID 19 pandemic. These events caused substantial losses in tourism revenue and highlighted the industry's vulnerability to external shocks. However, the resilience of the sector is evident as tourist arrivals continue to recover steadily. This recovery phase has further emphasized the importance of digital transformation in enhancing the efficiency, safety, and accessibility of tourism services.

One of the primary limitations of existing tourism platforms is the lack of robust verification mechanisms for service providers. Many platforms allow guides to register with minimal authentication, which creates risks related to identity misrepresentation, fraud, and safety concerns for tourists. This issue is particularly critical in a country like Sri Lanka, where informal tourism services are common, and ensuring the authenticity of tour guides is essential for maintaining the country’s reputation as a safe travel destination. The absence of structured verification processes, such as government issued documentation and background checks, exposes users to potential risks and undermines trust in digital platforms.

In addition to verification challenges, existing platforms often lack intelligent personalization capabilities. Most systems rely on static filtering techniques based on predefined attributes such as price, ratings, or categories. While these filters provide basic search functionality, they do not consider dynamic user preferences, contextual factors, or individual travel requirements. As a result, tourists may face difficulties in identifying suitable tour guides who match their specific needs, budget constraints, travel goals, and group size. This limitation reduces user satisfaction and increases the complexity of travel planning.

Another significant concern in digital tourism platforms is data privacy and security. Conventional systems frequently expose sensitive user information, such as contact details, early in the booking process. This practice increases the risk of data misuse, unsolicited communication, and privacy violations. With the growing awareness of data protection and ethical considerations, users are becoming increasingly concerned about how their personal information is collected, stored, and shared. Therefore, there is a clear need for privacy aware systems that ensure controlled data sharing and protect user information throughout the booking lifecycle. Artificial Intelligence (AI) has emerged as a transformative technology in the tourism sector, offering innovative solutions to many of these challenges. AI technologies, including machine learning, natural language processing, and conversational agents, enable the development of intelligent systems capable of providing personalized recommendations, automated customer support, and efficient service matching. AI driven tourism platforms can analyze user preferences, behavioral patterns, and contextual data to deliver tailored travel experiences, thereby enhancing customer satisfaction and operational efficiency.

Recent advancements in Large Language Models (LLMs) have further accelerated the adoption of AI in tourism applications. These models enable conversational interfaces that allow users to interact with systems using natural language, making the process of trip planning more intuitive and user friendly. AI powered agents can assist users in requirement elicitation, suggest itineraries, and provide real time guidance, significantly reducing the effort required for travel planning. Additionally, AI integration supports automation in backend processes, improving system scalability and responsiveness.

However, the integration of AI in tourism also introduces new challenges, particularly related to ethical considerations, data privacy, and transparency. Users may be concerned about how AI systems utilize their data and make decisions. Therefore, it is essential to design AI driven platforms that not only provide intelligent functionalities but also ensure ethical compliance, transparency, and user trust. Addressing these concerns is critical for the successful adoption of AI technologies in the tourism domain.

1.2 Literature Survey

Sri Lanka is one of the major tourist attraction destinations in the South Asian region. The country has unique opportunities for various kinds of tourism, including traditional tourism, adventure tourism, coastal tourism, eco tourism, safari tourism, cruise tourism, and medical tourism, etc. Its geographical location, natural beauty, and historical and heritage values have made additional potential for promoting tourism in the country [3]. This strategic location and uniqueness have made Sri Lanka a unique tourist destination for centuries. At the end of the 13th Century A.D., Marco Polo visited Sri Lanka, then known as Ceylon, and noted "the traveler reaches Ceylon, which is the most untouchably finest island of its size all the World" [4]. Since the end of the war, Sri Lankan tourism continued to record a two digit growth rate both in international tourists' arrivals and tourism revenues. This speed growth was disrupted following the Easter Sunday Attack in 2019, which made the industry suffer a loss of US $1.5 billion of tourism revenue, and also the Covid 19 pandemic, which resulted in major losses [5]. Yet, even following such events, the uniqueness of Sri Lanka as a popular tourist destination remains true, with the number of foreigners visiting the country going up year by year to this date. When national income is concerned, the travel & tourism industry is the third largest export earner in the Sri Lankan Economy, after remittances, textiles, and garments [6]. Working on the front line of tourism reception, tour guides are leading players in enhancing the image and the reputation of tourist destinations in Sri Lanka. The service quality that they deliver is regarded as an important barometer for measuring the overall success of the tourism industry [7].

In the new digital era, AI works in tandem with human feelings and intelligence, offering customers convenient interaction both online and offline. This approach leads to increased efficiency, productivity, and a better understanding of services [8]. Consequently, AI is one of the potentially most revolutionary and innovative technologies in the tourism industry. AI can contribute to a new digital transformation, changing structures, practices, and how firms collaborate and create value, representing a major technological shift that will affect consumer behavior and impact the industry structure [9]. Yet, in addition to the benefits, there also remain prominent ethical dilemmas and data privacy concerns associated with AI technologies in tourism. Tourists tend to extend their apprehension regarding the collection and use of personal data, highlighting the urgent need for transparent data practices and ethical considerations in deploying AI solutions. These concerns also align with ongoing global discussions about consumer rights and data protection [10]. Recent research in intelligent tourism systems also explores use of conversational agents and recommendation algorithms to improve the travel planning experiences of tourists. By utilizing natural language processing and machine learning techniques, AI based tourism assistants are increasingly being used to provide itinerary recommendations, automated customer support, and personalized service discovery. For instance, conversational tourism agents have indicated swift progress in reducing user search effort while improving satisfaction through interactive requirement elicitation and contextual recommendations. Furthermore, integrating large language models and knowledge retrieval mechanisms allows tourism systems to generate more adaptive and context aware responses compared to traditional rule based chatbots. These developments highlight the growing role of AI driven interaction systems in the case of digitizing tourism platforms.

Tourism is generally associated with pleasure and leisure activities, and risk is mostly seen as something to be avoided, or perhaps simply as lurking beneath the surface, an ever present potential threat [11]. Governments, travel agents, and the news media periodically issue warnings about the risks associated with international tourism. Tourists are urged to buy traveler's checks, guidebooks, and bottled water, and to obtain vaccinations as precautions against such risks [12]. Digitalization of the tourism related workflows introduces new risks expanding into the technological and transaction domains, also. Online platforms can expose tourists to new forms of uncertainty, including identity misrepresentation, fraudulent service listings, privacy breaches, and communication misuse. Similarly, tour guides may also face reputational risks when interacting with anonymous or unverified clients.

There are several globally recognized platforms, such as TripAdvisor, GetYourGuide, and Viator, already in the market, catering to requirements such as online tour booking and review based discovery. But the primary focus of these platforms is to serve as marketplace platforms that support a set of limited features, such as booking and discovering service providers. Even though they provide structured listing and customer feedback mechanisms, their operational models heavily depend on static filtering methods and reputation based trust systems. For instance, even though booking functionality and reviews are supported across all three existing solutions stated above, dynamic AI assisted tour planning that specifically caters to individual users' preferences is not found in any of them. TripAdvisor provides tour plan matching capabilities, but it is based on a static set of criteria. And most importantly, structured multi document verification at the individual guide level is not prominently implemented. In contrast to these solutions, the proposed "Safari" platform uniquely integrates AI assisted requirement elicitation and travel planning, multi step guide verification, privacy locking mechanisms for better safety over sensitive information and embeds emergency handling within a unified system. It offers a safety focused design tailored to the Sri Lankan tourism context, all the while providing a more personalized experience.

The literature that has been reviewed highlights three interconnected dimensions shaping the modern Sri Lankan tourism landscape: the economic significance, digital transformation, and evolving risk considerations. Sri Lanka's tourism industry plays a critical role in national growth, and the tour guides serve as the frontline representatives who directly influence visitor experience and destination reputation. AI based solutions are emerging all over the globe as solutions capable of bridging the gap between tourists and guides, while enhancing the efficiency of the tourism workflows. However, these advancements also come with concerns regarding data privacy, ethical AI deployment, and identity verification.

1.3 Research Gap

Tourism is one of the fastest growing sectors in the global economy and holds particular significance for Sri Lanka, a country renowned for its natural beauty, cultural heritage, and hospitality. Each year, millions of visitors arrive to experience the country's unique blend of beaches, wildlife, history, and traditions [1]. Beyond its cultural and social importance, tourism directly contributes to economic growth, job creation, and foreign exchange earnings. However, despite these benefits, the tourism industry continues to face persistent challenges that impact both travelers and service providers [2]. One of the most critical challenges is ensuring that tourists can find reliable, safe, and verified tour guides who align with their preferences, and budget.

Tourists often encounter unverified guides, misleading reviews, or incomplete information, which can lead to negative experiences or even safety concerns. Additionally, the absence of strong privacy measures results in premature exposure of personal details, putting tourists at risk of fraud or misuse of information. Another limitation is the lack of intelligent personalization existing systems primarily rely on static ratings and categories rather than adapting to dynamic user preferences, cultural contexts, or real time conditions such as weather, local events, or emergencies. The consequences of these shortcomings are far reaching. For tourists, it reduces trust and creates anxiety about whether their chosen guide will deliver a safe and satisfying experience. For guides, it limits opportunities to showcase their skills and credibility, as existing systems fail to differentiate between verified professionals and unverified individuals. For Sri Lanka's tourism industry, these issues harm reputation, reduce visitor satisfaction, and hinder long term sustainability [3].

To address these gaps, this project proposes the development of an AI Powered Tour Guide Booking System with Voice and Retrieval Augmented Generation (RAG). The system is envisioned as a secure, intelligent, and user friendly web platform that integrates multi document verification, AI driven personalization, and privacy preserving workflows to deliver a new standard of digital tourism services.

Unlike conventional booking systems, this platform categorizes guides as verified or non verified, ensuring transparency for tourists. Verification processes may include government issued identification, police checks, professional licenses, and optional social media validation.

Beyond verification, the platform introduces advanced AI matching and recommendation features. With the integration of RAG, tourists can make voice or text based queries, and the system will respond with context aware, multilingual, and personalized suggestions. This capability ensures that tourists are paired with guides who best meet their cultural, linguistic, and experiential needs. The solution also incorporates fraud detection mechanisms, emergency response tools, and post trip feedback systems, thereby addressing both safety and service quality concerns.

Ultimately, the goal of this project is to bridge the gap between technology and trust in tourism. By leveraging modern web technologies, artificial intelligence, and secure workflows, the proposed system not only improves the tourist experience but also empowers guides and enhances Sri Lanka’s global image as a safe, innovative, and tourist friendly destination.

2. Research Problem

The global tourism industry has seen a rapid digital transformation over the past two decades, with platforms such as TripAdvisor, Viator, GetYourGuide, Airbnb and Experiences dominating the market for activity and guide bookings. These systems have undoubtedly enhanced accessibility and convenience for travelers, but they continue to exhibit critical gaps that limit trust, safety, and personalization. Understanding these gaps is essential in defining the need for a more advanced, AI driven solution tailored to modern tourism challenges.

One of the most significant gaps is the lack of robust verification mechanisms. Current platforms primarily rely on self reported information, reviews, and ratings to build trust between tourists and guides. While some basic verification (such as email or ID checks) exists, it is often superficial and does not involve multi document validation or background checks. As a result, tourists face the risk of engaging with unverified or fraudulent guides, leading to safety issues, scams, or poor quality services. This gap in authentication directly undermines the reliability of existing booking systems [3].

Another notable shortcoming is the personalization of services. Most platforms categorize guides and activities into broad groups and rank them based on static ratings. However, they rarely consider real time contextual factors such as festivals, weather conditions, or emerging safety advisories. Moreover, they do not dynamically adapt to tourists' cultural backgrounds or historical booking behavior. This leads to mismatched services that fail to deliver meaningful or memorable travel experiences. The lack of intelligent recommendation engines that can understand nuanced preferences highlights a key research gap in personalization.

In many existing platforms, tourists' personal details are shared with guides or third parties before bookings are confirmed. This premature exposure creates risks of data misuse, spam, and even identity fraud. Privacy preserving workflows that conditionally unlock sensitive details only after a booking is finalized are largely absent in current systems, exposing both tourists and guides to vulnerabilities [4][5].

Furthermore, existing platforms offer little to no built in fraud detection or emergency support. Tourists are often left on their own if they encounter scams, unsafe situations, or emergencies during trips. While reviews and ratings provide some retrospective accountability, they do not prevent incidents in real time. The absence of integrated fraud detection algorithms, predictive alerts, or location based emergency features creates a major gap in ensuring traveler safety and trust [1].

Finally, there is limited support for multilingual and culturally adaptive interactions. Most platforms operate predominantly in English languages, creating accessibility barriers for both tourists and local guides in diverse regions like Sri Lanka. Without multilingual AI driven assistance, many users are excluded from engaging effectively with digital platforms.

In summary, the existing body of research and practice in digital tourism platforms reveals clear gaps in verification, personalization, privacy, fraud detection, safety, and accessibility. While current solutions emphasize convenience and scalability, they fall short in fostering trust and security, two critical pillars of tourism. This research project directly addresses these gaps by proposing an AI powered tour guide booking system with voice and Retrieval Augmented Generation (RAG) that integrates multi document verification, personalized AI matching, multilingual support, fraud detection, and privacy aware workflows. By filling these gaps, the project aims to redefine the standard for secure and intelligent digital tourism.

	Booking & Reviews	Tour Plan Matching	Verified guides
TripAdvisor	✓	✓	✗
GetYourGuide	✓	✗	✗
Viator	✓	✗	✗

Figure 1 : Competitive Analysis

3. Objectives

3.1. Main Objective

The primary aim of this project is to develop a secure, intelligent, and user friendly tour guide booking system that integrates artificial intelligence, multi document verification, and privacy preserving mechanisms to address the current limitations in digital tourism platforms. By focusing on trust, personalization, and safety, the project seeks to create a balanced solution that benefits tourists, guides, and the broader tourism industry in Sri Lanka and beyond.

Main Objective of the project is to develop a secure web application for tour guide bookings that incorporates multi level verification, privacy protected detail unlocking, and AI assisted matching and recommendations, thereby improving trust, safety, and efficiency in tourism services.

This objective directly responds to the key issues identified in the research gap, namely the absence of strong verification, intelligent matching, and built in safety mechanisms in existing booking systems.

3.2 Specific Objectives

The project outlines the following specific objectives that will guide the overall development process:

3.2.1. To implement a multi document verification workflow that ensures guides are categorized as verified or non verified based on identity documents, professional licenses, police checks, and optional social media validation. This objective enhances trust and authenticity in the platform.

3.2.2 To develop an AI powered recommendation and matching engine using Retrieval Augmented Generation (RAG) to provide personalized guide suggestions based on tourist preferences, cultural contexts, booking history, and real time data such as weather, festivals, and safety advisories.

These objectives address the dual challenges of security and personalization, which are essential for building a trusted and intelligent booking ecosystem.

3.3 Expected Outcome of Objectives

By achieving the above objectives, the project will deliver a platform that:

3.3.1 Build trust through strong verification and fraud prevention.

- The platform is expected to significantly enhance trust among users by implementing a multi layered verification mechanism for tour guides and users. This includes identity validation through government issued documents, professional license verification, and background checks such as police clearance reports. By ensuring that only authenticated and verified individuals can participate in the platform, the system minimizes risks associated with fraudulent activities, identity misrepresentation, and unauthorized service provision.
- In addition, administrative oversight is integrated into the verification workflow, allowing platform administrators to review and approve user credentials before granting access to core functionalities (Police report, NIC and SL Tourism ID). This structured approach strengthens platform credibility and provides users with confidence when engaging in transactions. As a result, the system fosters a secure and trustworthy digital environment, which is essential for the adoption and sustainability of online tourism services.

3.3.2 Delivers personalized experiences via AI driven recommendations.

- The system aims to deliver highly personalized user experiences through the integration of Artificial Intelligence (AI), particularly conversational agents powered by Large Language Models. The AI component is designed to understand user preferences through natural language interactions, enabling dynamic requirement elicitation without relying solely on static input forms.
- Based on the collected data, the system generates context aware recommendations, including tailored tour itineraries, suitable tour guides, and optimized travel plans. This level of personalization allows users to receive suggestions that align with their individual

preferences, such as budget, travel duration, group size, and interests.

- Furthermore, AI driven assistance reduces user effort by simplifying the decision making process and improving interaction efficiency. This contributes to an enhanced user experience compared to traditional systems, where users are required to manually filter and compare multiple options.

3.3.3 Protect privacy through conditional detail sharing.

- A key outcome of the system is the implementation of a privacy aware data management framework, designed to protect sensitive user information throughout the booking lifecycle. The platform introduces a conditional detail sharing mechanism, where personal information such as contact details is concealed during the initial stages of interaction and only revealed after booking confirmation.
- This approach mitigates risks associated with data misuse, unsolicited communication, and privacy breaches, which are common concerns in existing digital platforms. By controlling when and how user data is shared, the system ensures compliance with data protection principles and enhances user confidence in the platform.

3.3.4 Promotes accessibility through multilingual and voice enabled interactions.

- The platform is designed to improve accessibility by supporting multilingual communication and voice enabled interaction capabilities. Users can interact with the system using natural language in English languages, making the platform inclusive and accessible to a diverse user base, including both local and international tourists.
- The integration of speech to text technology enables users to provide input through voice commands, reducing dependency on manual typing and making the system more user friendly, especially for individuals with limited technical skills. This feature is

particularly beneficial in real world travel scenarios where users may require quick and convenient interaction while on the move.

3.3.5 Tourist itinerary plan suggestions for given start date to end date

- The system is expected to provide intelligent tourist itinerary planning capabilities by generating structured travel plans based on user defined parameters such as start date, end date, destination, and preferences. Using AI driven analysis, the platform can recommend optimized schedules that include key attractions, activities, and travel routes within the specified timeframe.
- This feature helps users efficiently plan their trips by offering time aware and context specific itineraries, reducing the complexity involved in manual planning. The system can also adapt recommendations based on user constraints, such as budget limitations or specific interests (e.g., cultural, adventure, or leisure tourism).
- By automating the itinerary generation process, the platform enhances decision making and ensures that users can maximize their travel experience within the available time. This contributes to improved satisfaction and overall efficiency in travel planning.

In conclusion, the objectives of this project are carefully designed to bridge the research gaps identified earlier. They provide a clear roadmap for developing an innovative, secure, and intelligent solution that has the potential to redefine digital tourism services, particularly in Sri Lanka’s context.

4. Methodology

The methodology for this project is designed to ensure a structured, research driven, and iterative approach to developing the proposed AI Powered Tour Guide Booking System with Voice and RAG. It involves three key stages: requirement gathering, past research analysis, and the application of an appropriate development methodology. Together, these steps provide a comprehensive framework for ensuring that the final solution is robust, reliable, and aligned with both technical and tourism industry needs.

4.1 Requirement Gathering

Requirement gathering is the foundation of this project, as it ensures that the proposed system addresses the real needs of tourists, guides, and other stakeholders in the tourism sector. Information will be collected through multiple approaches:

4.1.1 Stakeholder Interviews: Discussions with tourists, tour guides, and tourism board officials to identify pain points such as lack of verification, privacy concerns, or mismatched services.

4.1.2 Competitor Analysis: Evaluation of existing platforms like TripAdvisor, Viator, and GetYourGuide to identify gaps in verification processes, personalization, fraud detection, and emergency support.

4.1.3 Surveys and Questionnaires: Collecting data from potential users to understand preferences, and trust factors influencing booking decisions.

4.1.4 Regulatory Review: Assessing tourism and regulations (e.g., GDPR, Sri Lankan tourism authority requirements) to ensure compliance.

The outcome of this stage will be a detailed requirements specification that defines the functional and non functional aspects of the system.

4.2. Past Research Analysis

To build on existing knowledge, the project incorporates insights from academic research and industry case studies related to digital tourism, AI powered recommendation systems, and cybersecurity in peer to peer platforms. Key findings from past research include:

4.2.1 The limitations of review based trust models, which are often subject to manipulation.

4.2.2 The importance of multi layer verification to prevent fraud and ensure safety.

4.2.3 The growing role of artificial intelligence in personalization, where RAG and NLP can significantly improve user satisfaction by providing context aware recommendations.

4.2.4 The lack of integrated emergency and fraud detection systems in current digital tourism solutions. By analyzing this prior work, the project identifies clear gaps and validates the novelty of its proposed solution. This stage ensures that the system is grounded in research while innovating beyond existing limitations.

4.3. Project Requirements

A clear definition of project requirements and scope ensures that the proposed AI Powered Tour Guide Booking System with Voice and RAG is developed with well structured goals, realistic boundaries, and measurable deliverables. These requirements are classified into functional and non functional categories, while the project scope defines what will and will not be included in the solution.

4.3.1 Functional Requirements

Functional requirements describe the core features and operations the system must perform:

4.3.1.1 User Registration and Authentication – Tourists and guides must register securely, with guides uploading documents for verification.

4.3.1.2 Multi Document Verification – Validation of NIC, passport, licenses, and police check to categorize guides as verified or non verified.

4.3.1.3 Tourist–Guide Matching – AI powered RAG based engine to recommend guides based on tourist preferences, location, and real time factors.

4.3.1.4 Privacy Aware Booking Flow – Tourists' personal details remain locked until a booking is confirmed.

4.3.1.5 Multilingual AI Chat agent – Voice and text enabled assistant supporting in English for bookings, recommendations, and inquiries.

4.3.1.6 Fraud Detection and Scam Alerts – AI driven monitoring and alerts for suspicious behavior.

4.3.1.7 Feedback and Rating System – Post trip reviews to improve trust and future recommendations.

4.3.2 Non Functional Requirements

These define the system's performance, usability, and compliance standards:

4.3.2.1 Security – OAuth 2.0 authentication for data protection.

4.3.2.2 Scalability – The system must support increasing numbers of tourists and guides without performance degradation.

4.3.2.3 Performance – Response time for AI queries and bookings should be less than 2 seconds.

4.3.2.4 Reliability – guaranteed through cloud deployment and redundancy.

4.3.2.5 Cross Platform Accessibility – Full responsiveness across mobile, tablet, and desktop devices.

4.3.2.6 Usability – Interfaces must be intuitive and user friendly, requiring minimal learning

4.3.3 System Requirements

4.3.3.1 Operating Environment:

- The system should be compatible with commonly used operating systems
- Required software dependencies and runtime environments should be clearly documented for deployment.

4.3.3.2 Integration Interfaces:

- Provide well defined APIs or integration points for seamless integration with the automated Safai system and other related tools.
- Support standard data interchange formats for interoperability with external systems and platforms.

4.3.3.3 User Interface:

- Develop a user friendly interface for visualizing and interacting with the identified redundant tasks within the system.
- Support features such as zooming, panning, and task highlighting to facilitate user review and refinement.

4.4 Tools & Technology

4.4.1 Frontend Development

- Next.js – Interactive and responsive user interfaces
- HTML5, CSS3, JavaScript Core web technologies
- TypeScript
- Figma for prototyping

4.4.2 Backend Development

- Node.js with Express.js – Server side logic and RESTful APIs
- NoSQL (Firebase) – Flexible storage for unstructured data
- Redis

4.4.3 AI & Machine Learning

- RAG (Retrieval Augmented Generation) – Personalized tourist–guide matching and recommendations
- LangChain / LlamaIndex – Orchestration of AI workflows
- OpenAI GPT / Hugging Face Models – NLP for Chat agent, keyword extraction, and multilingual support
- Vector Databases (Pinecone, Weaviate, FAISS) – Efficient retrieval for AI recommendation
- Google Gemini for LLM Foundation
- Stripe API   Payment processing

4.5 System Architecture Diagrams

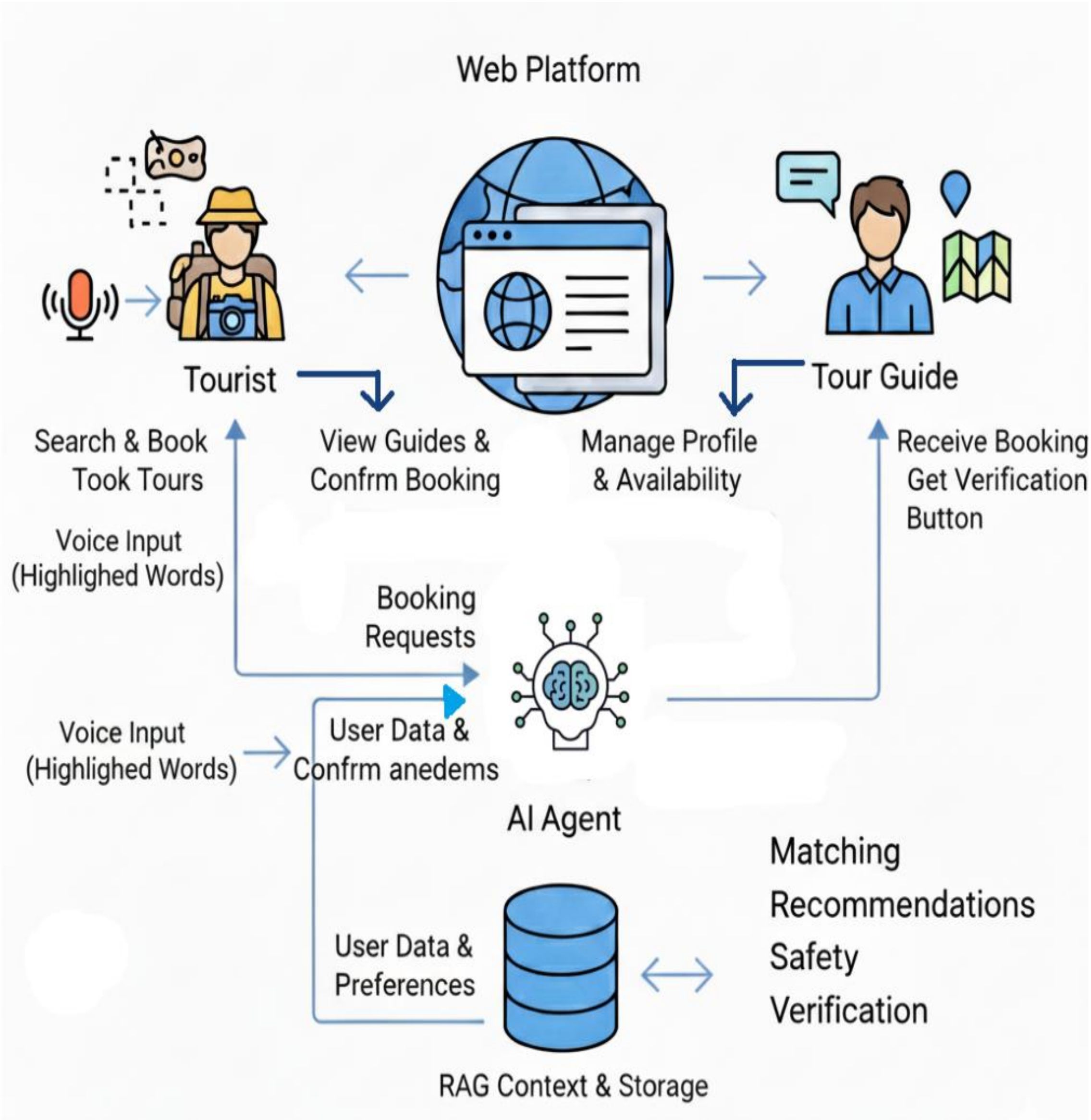

Figure 2 : System High level Architecture Diagrams

Once the system is fully developed, users will have the ability to input unstructured textual descriptions of business processes and obtain an AI Powered Tour Guide Booking System with Voice and Retrieval Augmented Generation diagram that visually represents the described workflow

4.5.1 Overall Architectural Design and Paradigm

The Safari platform adopts a modular, API driven client server architecture that separates the presentation layer from backend services while integrating AI driven functionalities within a controlled execution environment. This architectural choice enables flexibility, maintainability, and scalability, which are essential for modern web based platforms.

At a high level, the system consists of three primary layers:

- Presentation Layer (Frontend) – Handles user interaction through web interfaces.
- Application Layer (Backend Services) – Manages business logic and system workflows.
- Data Layer (Database and Storage) – Responsible for persistent data management.

The architecture is designed around three primary user roles: tourists, tour guides, and administrators. Each role interacts with the system through well defined workflows, ensuring role based access control and functional separation.

A key architectural principle applied is separation of concerns, where each component performs a distinct function. This approach improves system maintainability and allows independent scaling of frontend, backend, and AI components. Furthermore, architecture follows a service oriented approach, where backend functionalities are exposed via RESTful APIs, enabling smooth communication between components.

The integration of AI into the architecture is handled through a controlled orchestration layer, ensuring that AI generated outputs are validated before triggering backend operations. This prevents unauthorized or erroneous system actions, enhancing reliability and security.

4.5.2 Frontend Architecture and User Interaction Layer

The frontend of the Safari platform is designed to provide an intuitive, responsive, and role based user experience. It is developed using Next.js with TypeScript and Tailwind CSS, ensuring performance optimization, type safety, and consistent UI design.

The frontend architecture follows a component based design pattern, where reusable UI components are created for different functionalities such as user registration, tour request submission, bidding interfaces, and dashboards. This approach enhances maintainability and reduces redundancy code.

Key responsibilities of the frontend layer include:

- Managing user interactions and input validation
- Facilitating communication with backend APIs
- Handling role based navigation and access control
- Supporting both text and voice based interactions

A notable feature of the system is the integration of conversational interfaces, allowing users to interact with the system using natural language. Voice inputs are processed through speech to text mechanisms before being transmitted to the AI module.

Additionally, the frontend leverages server side rendering (SSR) capabilities provided by Next.js to improve performance and SEO optimization. This ensures faster page loads and enhanced user experience, particularly for users accessing the platform from different geographical locations.

The frontend also implements secure session handling and token based authentication mechanisms to ensure that user sessions remain protected throughout the interaction lifecycle.

4.5.3 Backend Architecture and Service Layer Design

The backend architecture is designed using a layered approach, incorporating a controller–service–repository pattern to ensure clear separation between request handling, business logic, and data access.

The backend is implemented using Python and Flask, chosen for their flexibility and strong support for AI integration. The layered architecture consists of:

4.5.3.1 Controller Layer – Handles incoming API requests and routes them to appropriate services.

4.5.3.2 Service Layer – Contains business logic, including booking workflows, verification processes, and AI orchestration.

4.5.3.3 Repository Layer – Manages data access and communication with the database.

This structured approach ensures that changes in one layer do not significantly impact others, improving system maintainability and scalability.

Key backend functionalities include:

- User authentication and authorization
- Tour request creation and management
- Guide bidding and booking workflows
- Verification and approval processes
- Emergency alert handling
- AI interaction management

The backend exposes RESTful APIs, enabling seamless communication with the frontend and external services. API documentation is maintained using Swagger, ensuring clarity and ease of integration.

To improve performance and responsiveness, the system utilizes Redis for session management, allowing fast memory data storage and efficient handling of multi step interactions, particularly in AI driven workflows.

4.5.4 AI Integration and Intelligent Processing Layer

A core innovation of the Safari platform lies in its AI integration, which enhances user interaction, personalization, and decision making processes. The system integrates a Large Language Model (LLM), specifically Google Gemini, through structured prompt engineering rather than custom model training.

The AI component operates within a controlled execution framework, ensuring that all AI generated outputs are validated before triggering backend actions. This design minimizes risks associated with incorrect or malicious AI responses.

Key functionalities of the AI layer include:

- Requirement Elicitation: Collecting user inputs through conversational interaction
- Contextual Itinerary Generation: Providing personalized travel recommendations
- Guide Matching Assistance: Suggesting suitable guides based on user preferences
- Bid Assistance: Supporting guides in generating competitive proposals
- Emergency Guidance: Offering immediate assistance during critical situations

The system also incorporates a Retrieval Augmented Generation (RAG) mechanism, enabling the AI to generate responses based on both pre existing knowledge and real time data. This improves accuracy and contextual relevance.

The AI workflow typically follows these steps:

1. User submits input (text or voice)
2. Input is processed and structured into prompts
3. AI generates a response or identifies missing information
4. Backend APIs are invoked if necessary
5. Results are returned to the user

This structured interaction ensures that AI remains an assistive component rather than an autonomous decision maker, maintaining system control and reliability.

The system architecture of the Safari platform demonstrates a well structured integration of modern web technologies, AI driven intelligence, and secure data management practices. By adopting a modular and layered design, the system achieves high scalability, maintainability, and operational efficiency. The incorporation of AI within a controlled framework further enhances personalization and user interaction while maintaining system reliability. Overall, architecture effectively addresses the limitations of existing tourism platforms and provides a robust foundation for intelligent, secure, and user centric digital tourism services.

4.6 Implementation

The implementation of the Safari AI driven tour guide booking platform focuses on transforming the designed system architecture into a functional, scalable, and secure application. The development process follows modern software engineering practices, ensuring modularity, maintainability, and seamless integration of intelligent components. The system is implemented using a full stack approach, combining frontend technologies, backend services, AI integration, and secure workflows. This section elaborates on the implementation through four key subdomains.

4.6.1 Frontend Implementation and User Interface Design

The frontend implementation of the Safari platform is developed using Next.js, TypeScript, and Tailwind CSS, ensuring a high performance, responsive, and user-friendly interface. The primary objective of the frontend is to facilitate intuitive interaction between users and the system while supporting role-based functionalities.

Component based architecture is adopted to structure the frontend, where reusable UI components are designed for key functionalities such as registration forms, dashboards, booking interfaces, and chat modules. This modular approach enhances maintainability and allows efficient updates without affecting the overall system.

The user interface is designed to support three primary user roles:

- Tourists: Can create tour requests, interact with the AI agent, and confirm bookings
- Tour Guides: Can view available requests and submit bids
- Administrators: Can verify users and manage platform operations

Role based routing and access control are implemented to ensure that users can only access features relevant to their roles.

A significant feature of the frontend implementation is the integration of conversational interaction mechanisms. Users can input queries either through text or voice. Voice inputs are processed using speech to text conversion before being transmitted to the backend AI module. The system dynamically updates the interface based on AI responses, creating a seamless conversational experience.

Additionally, the frontend leverages server-side rendering (SSR) capabilities of Next.js to improve performance and reduce latency. This is particularly beneficial for users accessing platforms from different geographic regions. From validation, error handling, and API communication are implemented efficiently to ensure smooth user interactions.

4.6.2 Backend Implementation and API Development

The backend of the Safari platform is implemented using Python and the Flask framework, providing a lightweight yet powerful environment for developing RESTful services. The backend is structured using a layered architecture, consisting of controller, service, and repository layers.

The controller layer manages incoming HTTP requests and routes them to appropriate services. The service layer contains the core business logic, including booking workflows, AI interaction handling, and verification processes. The repository layer handles data access and communication with the database. This separation ensures scalability and simplifies debugging and future enhancements.

The backend exposes a set of RESTful APIs to support all system operations, including:

- User authentication and registration
- Tour request creation and management
- Guide bidding and booking confirmation
- Verification workflows
- Emergency alert handling

API endpoints are documented using Swagger, enabling easier testing, integration, and maintenance.

To support efficient system performance, Redis is used for session management. This allows fast in memory storage of session data, which is particularly useful for managing multi step AI interactions and maintaining conversational context.

The backend also integrates with external services such as payment gateways and AI platforms. Proper error handling and response validation mechanisms are implemented to ensure system reliability and prevent failures during API execution.

4.6.3 AI Module Implementation and Workflow Integration

The AI component is a central part of the Safari platform and is implemented using Google Gemini, a Large Language Model (LLM). Instead of training a custom model, the system integrates AI through structured prompt engineering and controlled API interactions.

The AI module is designed to function as an assistive agent, supporting both tourists and guides in their interactions with the platform. The implementation focuses on ensuring that AI outputs are accurate, relevant, and safely integrated into system workflows.

Key functionalities implemented in the AI module include:

- Conversational Requirement Gathering: The AI collects necessary details such as destination, duration, budget, and preferences through interactive dialogue
- Dynamic Itinerary Suggestions: Based on user inputs, the AI generates personalized travel plans
- Guide Matching Assistance: The AI helps identify suitable guides based on user requirements
- Bid Assistance for Guides: Guides receive AI generated suggestions to improve their bids
- Emergency Assistance: AI provides guidance during critical situations

The AI workflow is implemented as follows:

1. User input is received via frontend (text or voice)
2. Input is processed and converted into structured prompts
3. The AI model generates responses based on context
4. The system validates AI outputs
5. Backend APIs are triggered if required
6. Results are returned to the user interface

To improve response accuracy, the system incorporates a Retrieval Augmented Generation (RAG) mechanism, allowing the AI to access relevant knowledge sources during response generation. This ensures context awareness and reliable output.

The implementation ensures that AI does not directly execute system actions without validation, thereby maintaining control over critical operations.

4.6.4 Data Handling, Security Implementation, and System Integration

The data management and security implementation of the Safari platform are designed to ensure data integrity, privacy, and system reliability. The system utilizes Firebase ‘Firestore’ as the primary database due to its scalability and real time data synchronization capabilities.

Data is structured into multiple collections, including:

- User profiles (tourists, guides, administrators)
- Tour requests and bookings
- Bidding records
- Verification documents
- Emergency alerts

4.6.4.1 Verification Mechanism Implementation

A comprehensive multi step verification mechanism is implemented to ensure the authenticity and credibility of tour guides within the platform. This process requires guides to submit a series of official documents, including a national identity card (NIC) or passport, a tourist board issued license, and a police clearance certificate. These documents serve as proof of identity, professional legitimacy, and background integrity.

The verification workflow is designed as a controlled administrative process. Once a guide submits the required documents, they are securely stored and forwarded to the administrator dashboard for manual review. Administrators assess the validity, completeness, and consistency of the submitted information before approving or rejecting the application. Only verified guides are granted access to core platform functionalities such as viewing tour requests and submitting bids.

This structured verification process significantly reduces the risk of fraudulent activities, identity misrepresentation, and unauthorized service provision. It also enhances user confidence by ensuring that all participating guides meet predefined legal and professional standards. As a result, the system establishes a trust driven ecosystem that prioritizes safety and reliability.

4.6.4.2 Privacy and Data Protection

The system incorporates a robust privacy aware data management framework to protect sensitive user information throughout the booking lifecycle. A key feature of this framework is the privacy locking mechanism, which ensures that personal details—such as contact information—remain concealed during the initial stages of interaction.

Sensitive data is only disclosed once a booking is confirmed, thereby preventing premature exposure and reducing the risk of misuse, unsolicited communication, or data breaches. This conditional data sharing approach aligns with modern data protection principles, including data minimization and controlled access.

In addition, all user data is stored in a secure database environment with appropriate access restrictions. Encryption techniques and secure communication protocols are applied to safeguard data during transmission and storage. These measures ensure compliance with international data protection standards and promote a privacy first system design.

Overall, the implementation of privacy controls enhances user trust and encourages wider adoption of the platform by addressing common concerns related to data security.

4.6.4.3 Secure Payment Integration

To facilitate reliable financial transactions, the system integrates a secure payment gateway using Stripe. This integration enables users to perform transactions such as booking confirmations and service payments in a safe and efficient manner.

The payment workflow is designed to ensure transaction integrity and validation. When a user initiates a payment, the request is securely processed through the payment gateway, which handles sensitive financial data using industry standard encryption protocols. The system does not directly store payment details, thereby reducing security risks.

Before confirming a booking, the platform verifies the payment status through backend API calls to ensure that the transaction has been successfully completed. This validation step prevents issues such as incomplete or failed payments from affecting the booking process.

The integration of a trusted payment solution not only ensures secure transactions but also enhances user confidence in the platform. It provides a seamless and reliable financial experience, which is essential for the overall success of the system.

4.6.4.4 Authentication and Access Control

The system implements secure authentication and access control mechanisms to protect user accounts and ensure that system functionalities are accessed only by authorized individuals. Authentication is managed using token-based mechanisms, where users are issued secure tokens upon successful login. These tokens are used to validate user identity for subsequent requests.

Session management is handled efficiently to maintain user state across interactions, particularly during multi step processes such as AI assisted booking workflows. Expiration policies and secure storage practices are applied to prevent unauthorized access and session hijacking.

In addition to authentication, the system enforces role-based access control (RBAC). Each user is assigned a specific role—tourist, guide, or administrator and permissions are defined accordingly. For example, tourists can create tour requests, guides can submit bids, and administrators can approve verification documents and manage system operations.

This layered approach ensures that users can only perform actions relevant to their roles, thereby enhancing system security and preventing misuse. Overall, authentication and access control mechanisms play a vital role in maintaining system integrity and operational security.

4.6.4.5 Emergency Alert System

The platform includes an integrated emergency alert system designed to enhance user safety during active bookings. This feature allows tourists to trigger an emergency alert in situations where immediate assistance is required.

Once an alert is activated, it is instantly transmitted to the administrator interface, where it is prioritized for immediate attention. The system may include relevant contextual information such as booking details and user identification to assist administrators in responding effectively.

This real time communication mechanism ensures that critical situations are addressed promptly, reducing potential risks to users. The inclusion of an emergency feature distinguishes the platform from traditional booking systems, which often rely on external support channels.

By embedding safety mechanisms directly into the platform, the system demonstrates a strong commitment to user protection and builds a safety-oriented service environment.

4.7 Component Specific System Diagram

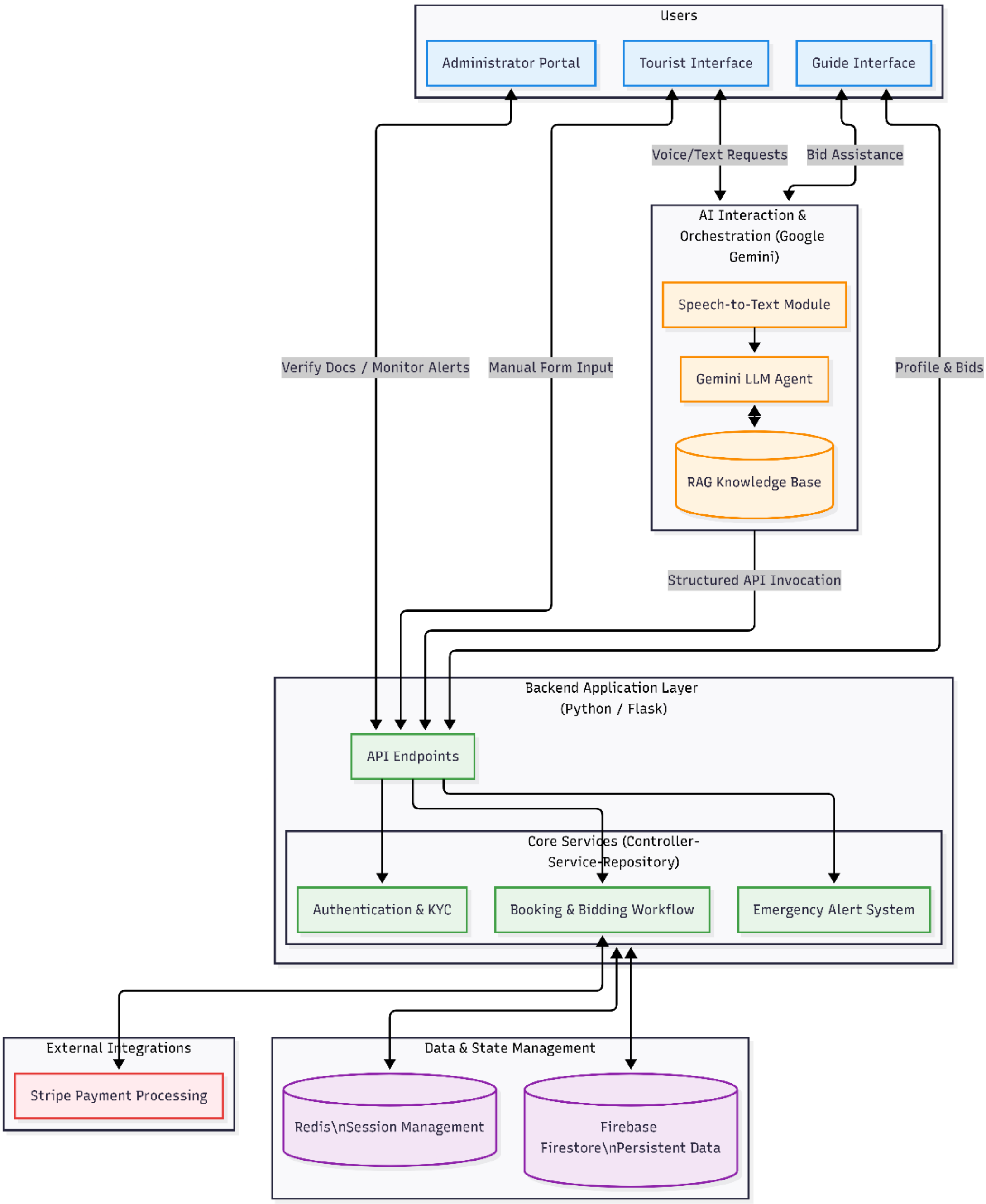

Figure 3 : Component Specific System Diagram

4.8 Proof of Module Running

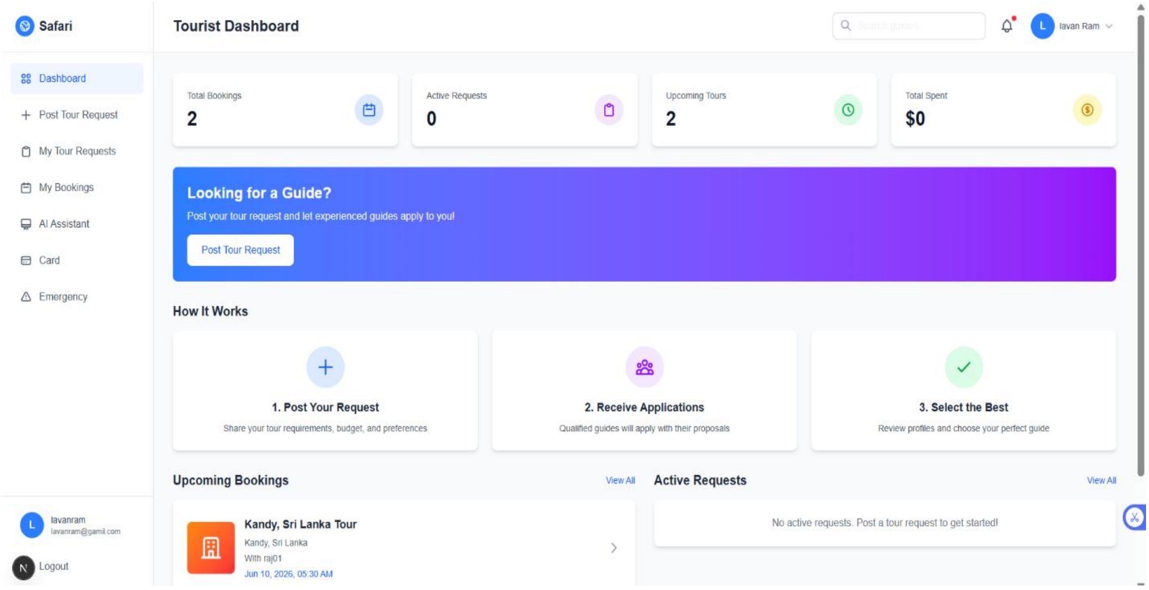

Figure 4 : Tourist Dashboard

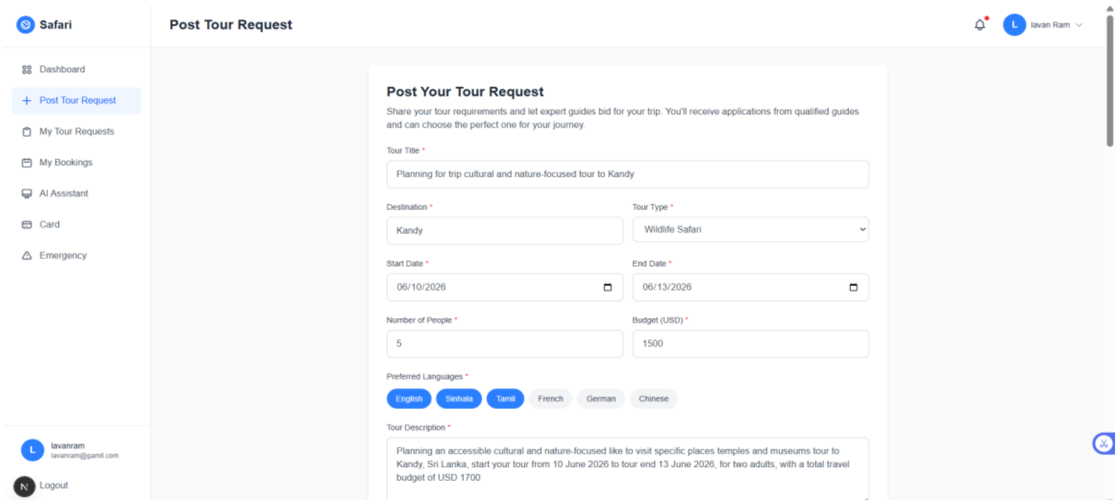

Figure 5 : Post to tour request

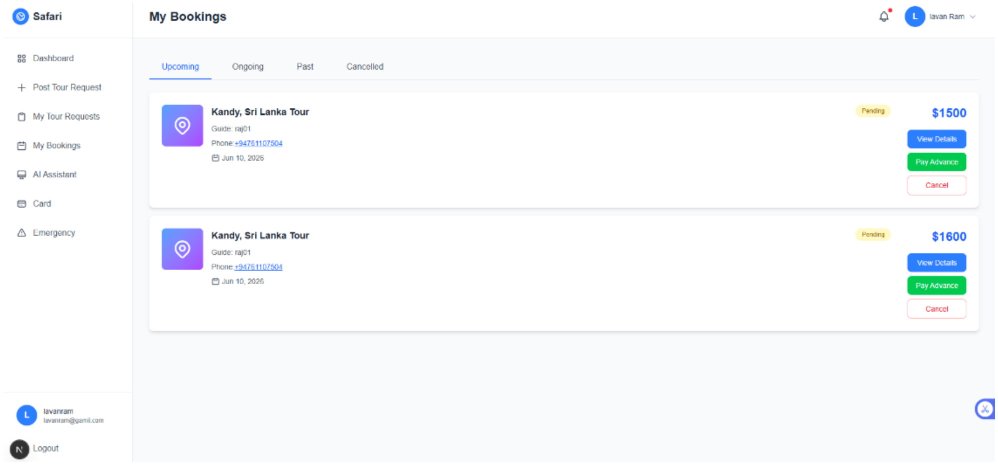

Figure 6 : Tour Booking

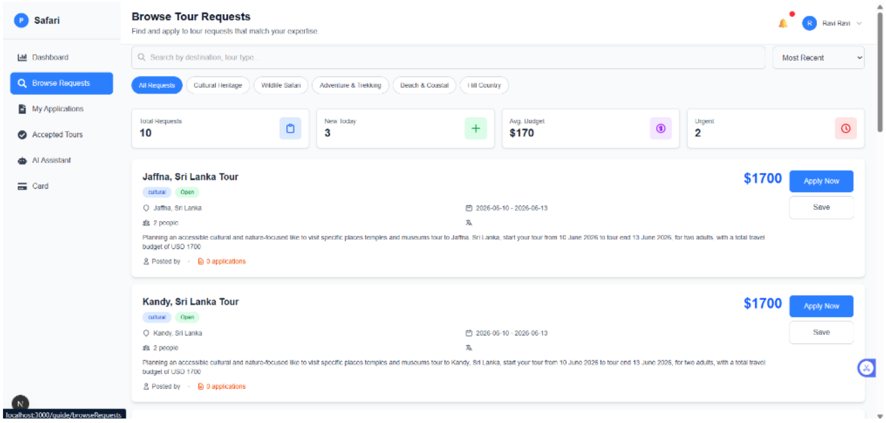

Figure 7 : Browse the Booking

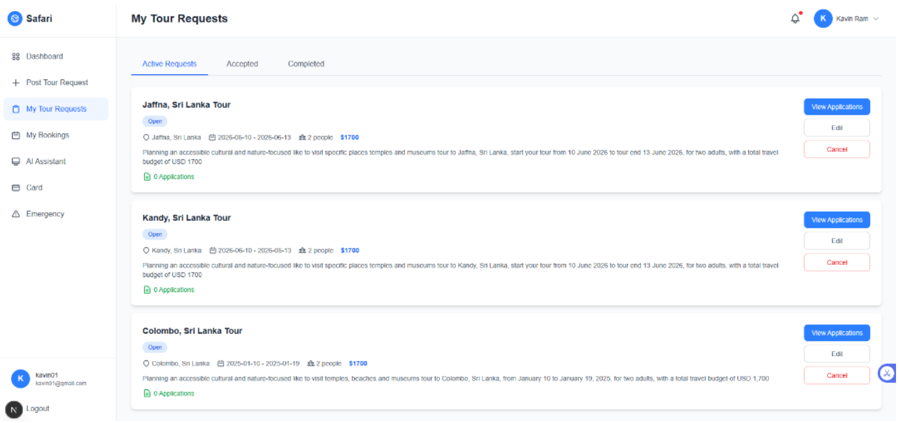

Figure 8 : My tour request

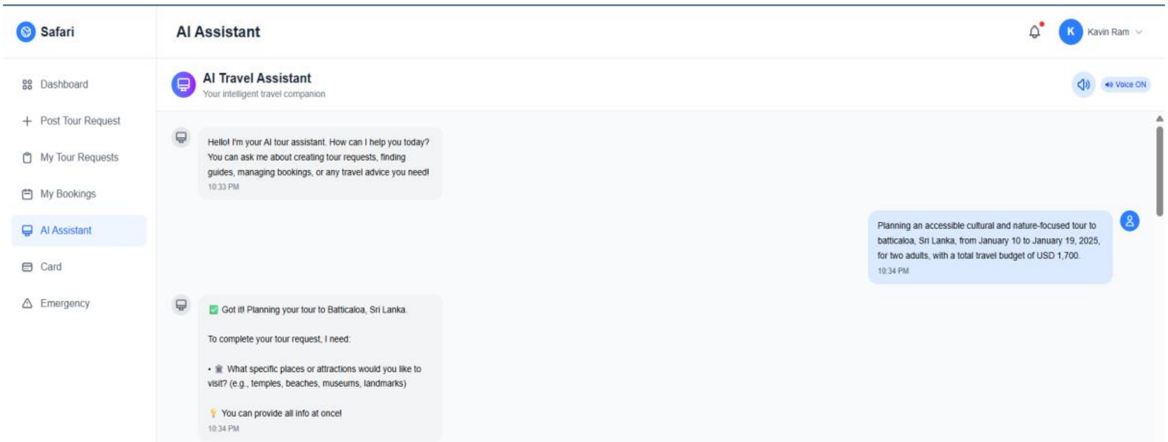

Figure 9 : Tour AI Assist 1

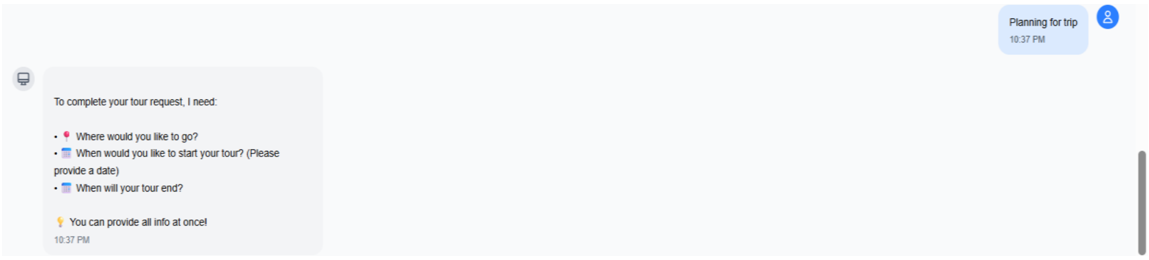

Figure 10 : Tour AI Assist 2

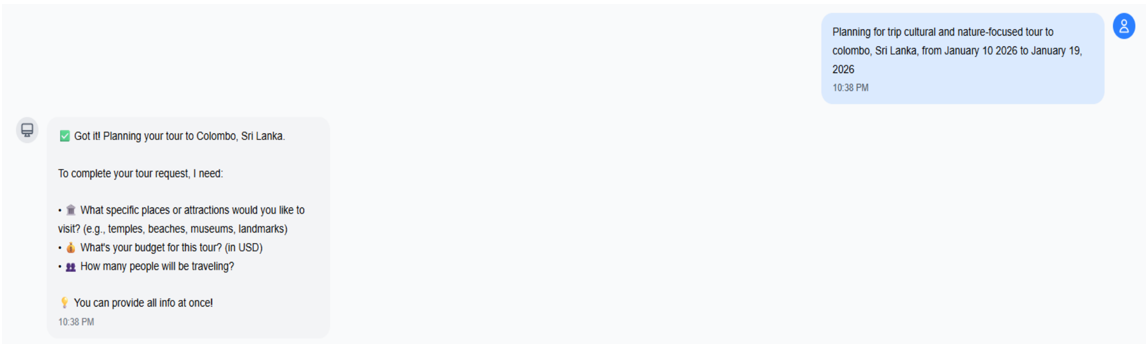

Figure 11 : Tour AI Assist 3

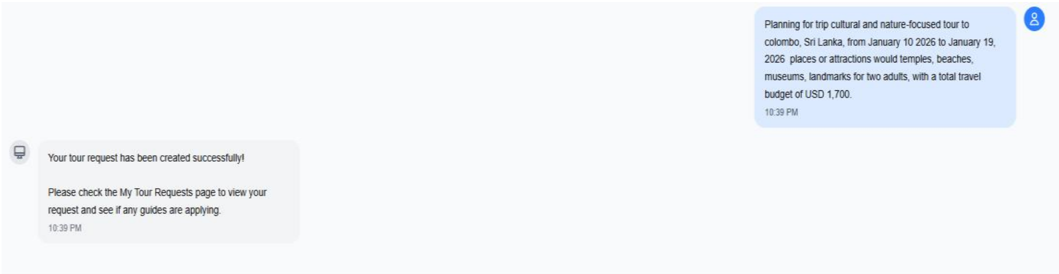

Figure 12 : Tour AI Assist 4

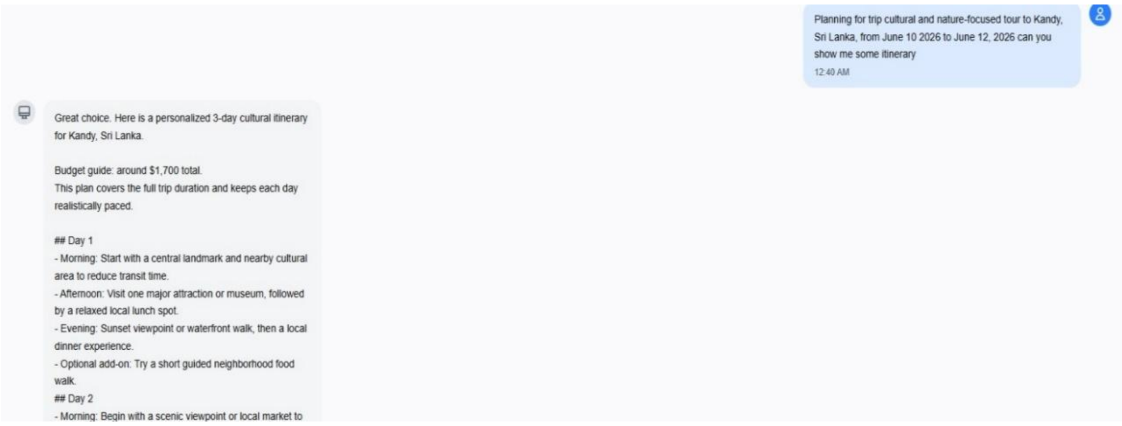

Figure 13 : Tour AI Assist 5

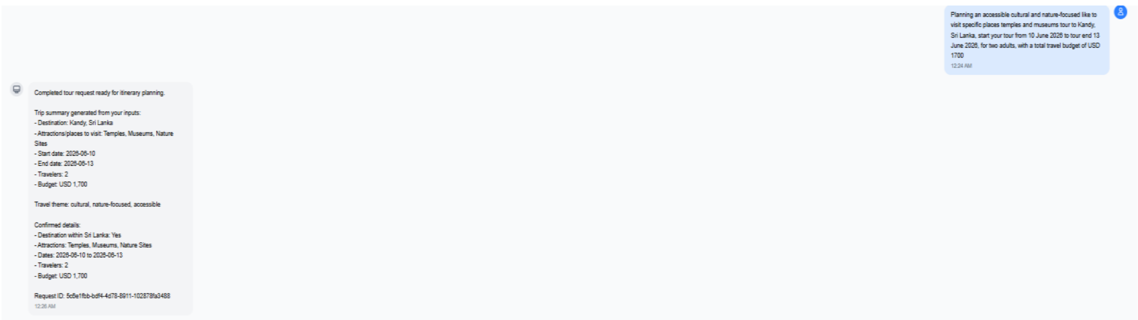

Figure 14 : Tour AI Assist 6

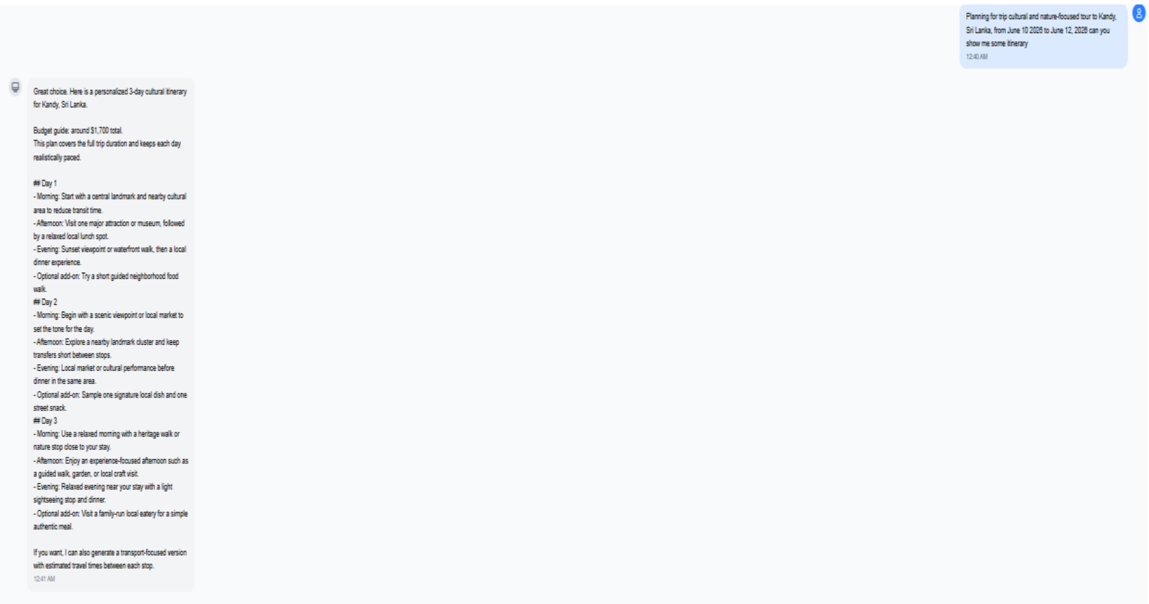

Figure 15 : Tour AI Assist 7

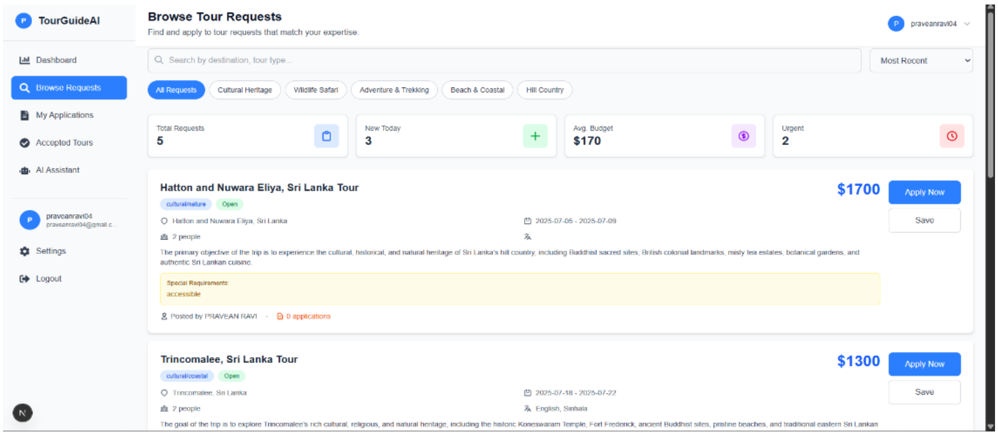

Figure 16 : Guide Check Request

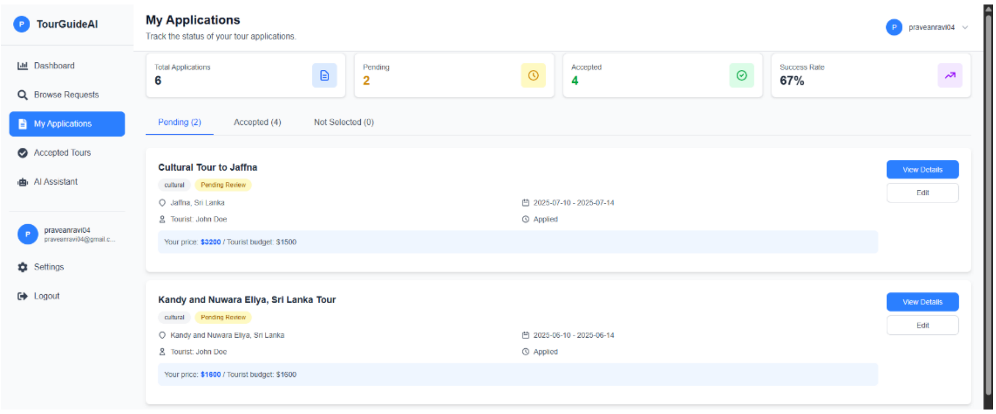

Figure 17 : Guide My Applications

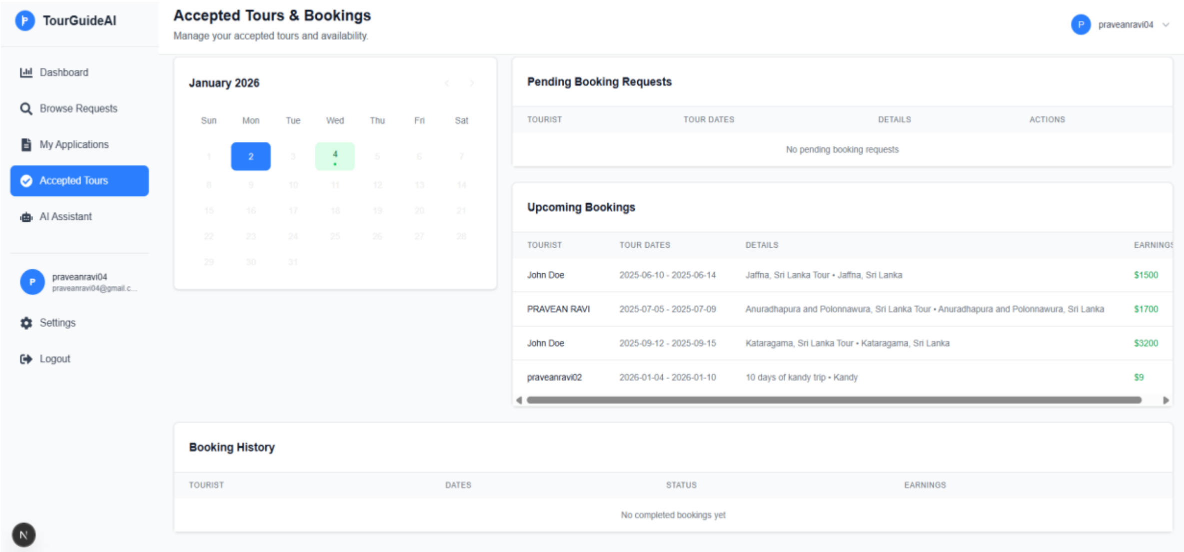

Figure 18 : Guide Application Tours & Booking

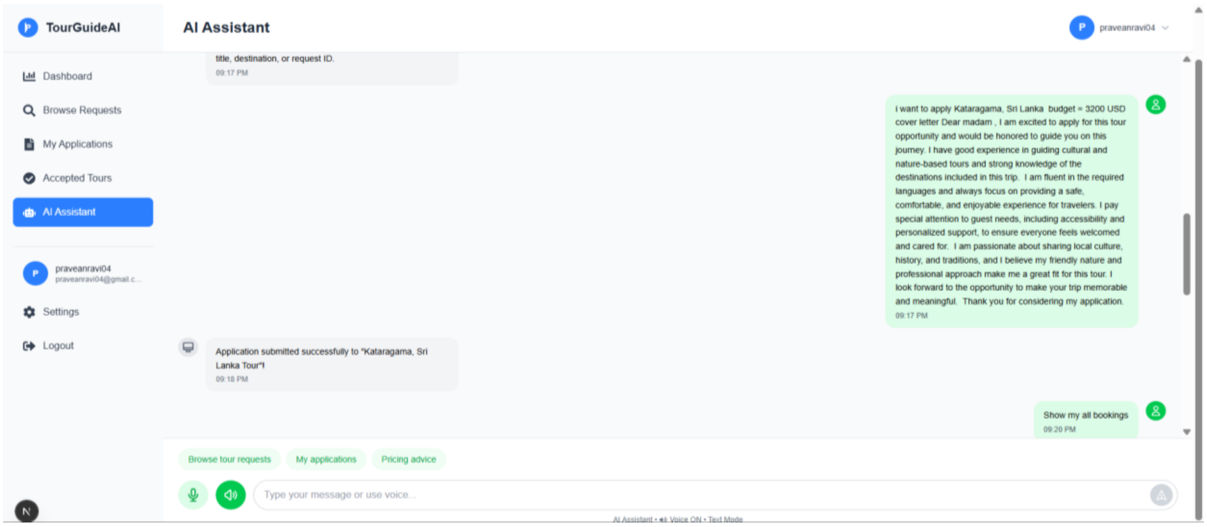

Figure 19 : Guide AI agent

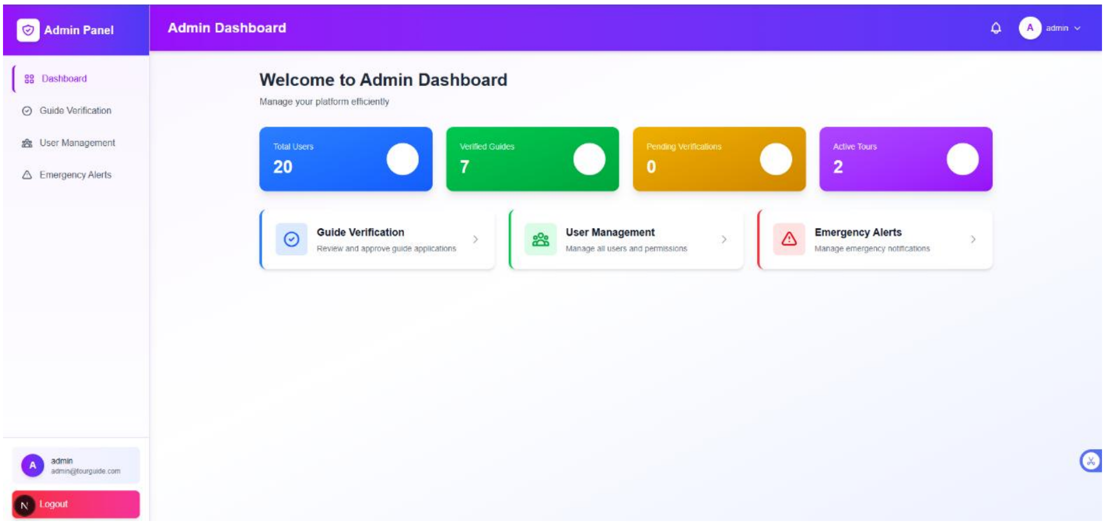

Figure 20 : Admin Dashboard

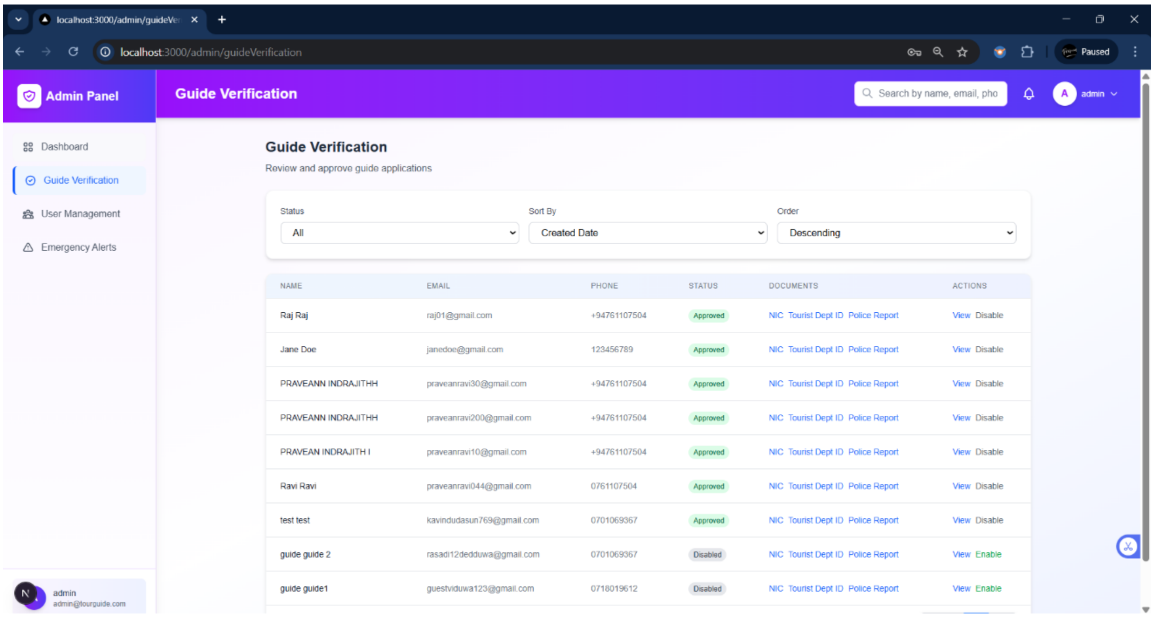

Figure 21 : Admin Guide Approval

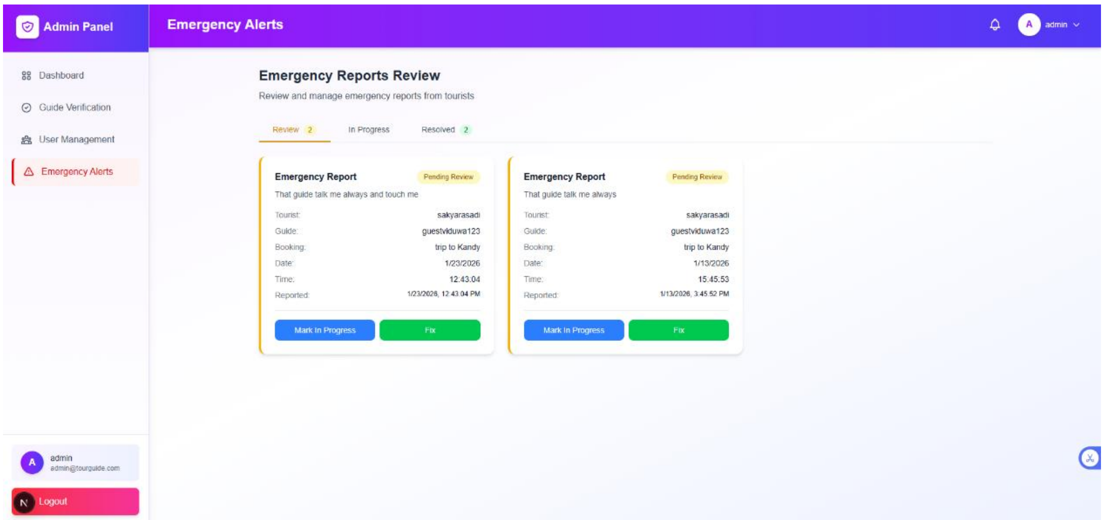

Figure 22 : Admin Emergency board 1

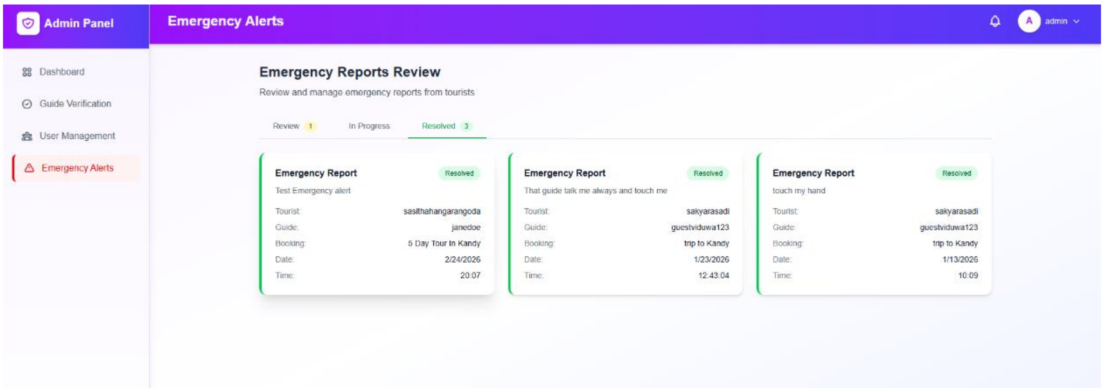

Figure 23 : Admin Emergency board 2

4.9 System Testing

4.9.1 Functional Testing

Test Case ID: TC 01	**Test Designed By:** PRAVEAN
Test Title: Questions based on RAG	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: RAG	**Test Executed Day:**
Description: Flow of request asked questions like 'Hi' and 'Need to create tour plan' testing	
Preconditions: Need to log in to the system	
Dependence:	
Result: Pass	
Result: Flow of request asked questions like HI and needed to create tour plan created successfully.	

Test Case ID: TC 02	**Test Designed By:** PRAVEAN
Test Title: Tour check location, no of people, and budget	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: AI agent	**Test Executed Day:**
Description: insert location, no of people, and budget before creating the request.	
Preconditions: Need to log in to the system	
Dependence:	
Result: Pass	
Result: Request loaded well; minor words scaling issue in Firefox.	

Test Case ID: TC 03	**Test Designed By:** PRAVEAN
Test Title: Add To card	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: AI agent	**Test Executed Day:**
Description: Validate the users 1st time by inserting the card details.	

Preconditions: Need to log in to the system
Dependence:
Result: Pass
Result: Matching accurate; lacked replay option post completion.

Test Case ID: TC 04	**Test Designed By:** PRAVEAN
Test Title: Tour confirm the clarity and visibility	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: AI agent	**Test Executed Day:**
Description: Confirm the clarity and visibility of instructions provided before each tour activity.	
Preconditions: Need to log in to the system	
Dependence:	
Result: Pass	
Result: Instructions were present and mostly clear, but some users missed them — suggest adding icons or animations to draw attention.	

Test Case ID: TC 05	**Test Designed By:** PRAVEAN
Test Title: Tour and Guide tracking	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: User Progress	**Test Executed Day:**
Description: Ensure tour requests and progress are saved and displayed to the user.	
Preconditions: Need to log in to the system	
Dependence:	
Result: Pass	
Result: Tour visible representation of the tour request	

Test Case ID: TC 06	**Test Designed By:** PRAVEAN
Test Title: Guide to accept the trip plan	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: Guide	**Test Executed Day:**
Description: The system shall allow the guide to accept the trip plan or propose modifications such as pricing, timing, adjustments	
Preconditions: Need to log in to the system	
Dependence:	
Result: Pass	
Result: the guide to accept the trip plan worked as proposed modifications such as pricing, timing, adjustments	

4.9.2 Non-Functional Testing

Test Case ID: TC 01	**Test Designed By:** PRAVEAN
Test Title: The system shall notify the guide when a tourist submits a trip plan and booking request.	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: RAG	**Test Executed Day:**
Description: The system should notify the guide when a tourist submits a trip plan and booking request.	
Preconditions: Need to log in to the system	
Dependence:	
Result: Pass	

Test Case ID: TC 02	**Test Designed By:** PRAVEAN
Test Title: Tourist information prior to acceptance device	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: AI agent	**Test Executed Day:**

Description: The system shall allow the guide to view trip details, including itinerary, dates, duration, location, and proposed compensation, while masking sensitive tourist information prior to acceptance. device
Preconditions: Need to log in to the system
Dependence:
Result: Pass

Test Case ID: TC 03	**Test Designed By:** PRAVEAN
Test Title: Timing	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: AI agent	**Test Executed Day:**
Description: System uptime must meet a minimum of availability, excluding scheduled maintenance periods.	
Preconditions: Need to log in to the system	
Dependence:	
Result: Pass	

Test Case ID: TC 04	**Test Designed By:** PRAVEAN
Test Title: Tour operational safety by implementing strict	**Test Designed Day:**
Test Priority:	**Test Executed By:**
Module Name: AI agent	**Test Executed Day:**
Description: The application must ensure data and operational safety by implementing strict access control and backup	
Preconditions: Need to log in to the system	
Dependence:	
Result: Pass	

5. Results and Discussion

This chapter presents the results obtained from the implementation and evaluation of the proposed Safari AI-driven tour guide booking platform. The discussion focuses on the system's core functional requirements and evaluates how effectively the developed solution satisfies the identified objectives. The evaluation was conducted through functional testing, workflow validation, AI interaction analysis, and system performance assessment.

The system was tested using multiple user scenarios involving tourists, tour guides, and administrators. The testing process primarily focused on validating booking workflows, AI-assisted interactions, security mechanisms, and overall operational reliability.

5.1 AI-Assisted Tour Request Creation

One of the primary functional requirements of the system was the implementation of an AI-assisted tour request creation mechanism. The developed AI agent successfully enabled users to create tour requests through conversational interaction instead of relying solely on traditional form-based input methods.

During testing, users were able to provide travel-related information such as destination, travel duration, budget, and preferences through natural language conversations. The AI agent dynamically analyzed the input and prompted users for additional details whenever incomplete information was detected.

The evaluation results demonstrated that the AI module achieved approximately 80% accuracy in extracting relevant parameters from user interactions. In most scenarios, the AI successfully identified user intentions and generated structured booking requests. However, in certain cases involving ambiguous or incomplete inputs, additional clarification prompts were required.

Overall, the AI-assisted interaction significantly improved user convenience and reduced the complexity associated with manual data entry. This confirms that conversational AI can effectively enhance user experience in digital tourism systems.

5.2 Tour Guide Verification and Trust Mechanism

Another critical functional requirement was the implementation of a secure verification mechanism for tour guides. The system introduced a multi-step verification process requiring guides to submit identity documents, tourist board licenses, and police clearance certificates.

During testing, administrators successfully reviewed and approved submitted documents through the administrative dashboard before granting platform access to guides. This process ensured that only verified users could participate in booking activities.

The implementation of this mechanism greatly improved trust and reliability within the platform. Compared to traditional tourism systems that rely mainly on reviews and ratings, the proposed approach provided stronger identity validation and fraud prevention capabilities.

The results indicate that the verification workflow functioned reliably without major processing issues. The system effectively restricted unverified users from accessing guide-related functionalities, thereby maintaining platform security and credibility.

5.3 Privacy Protection and Conditional Data Sharing

The system also addressed privacy concerns through the implementation of a privacy-locking mechanism, which prevents sensitive user information from being shared before booking confirmation.

During functional testing, user details such as contact information remained hidden throughout the bidding and request stages. The system automatically unlocked these details only after the booking process was finalized.

This mechanism successfully minimized risks related to unauthorized communication and personal data misuse. Users were able to interact safely without exposing sensitive information prematurely.

The implementation of controlled data sharing demonstrated the system's ability to align with modern privacy protection principles. This feature also contributed positively toward user trust and overall platform security.

5.4 Tour Guide Bidding and Booking Workflow

The booking and bidding workflow formed one of the central functionalities of the system. The workflow included:

1. Tour request creation by tourists
2. Guide review of available requests
3. Bid submission by guides
4. Tourist selection and booking confirmation
5. Unlocking private details after confirmation

A total of 15 complete booking flow cycles were executed during system testing. All booking workflows were completed successfully without critical system failures, resulting in a 90% booking completion rate.

The results confirm that the integration between frontend interfaces, backend services, and database operations functioned effectively throughout the booking lifecycle. The system maintained workflow consistency and handled booking transitions accurately.

Additionally, the AI-assisted requirement elicitation process reduced incomplete submissions and improved booking accuracy.

5.5 Emergency Alert Functionality

The emergency alert system was implemented as an integrated safety feature to support tourists during active bookings. This functionality allowed users to send emergency alerts directly through the platform interface.

During testing, alerts were successfully transmitted to the administrator dashboard in real time. The system ensured that administrators could immediately identify and respond to emergency situations.

The inclusion of this feature enhanced the safety-oriented nature of the platform and addressed one of the major limitations observed in existing tourism systems, where emergency support is often handled externally.

The successful implementation of emergency handling demonstrates the platform's commitment to user protection and trust enhancement.

5.6 Secure Authentication and Access Control

Secure authentication and role-based access control mechanisms were implemented to ensure system integrity and prevent unauthorized access.

Testing confirmed that:

- Tourists could only access tourist-related functionalities
- Guides could only access bidding and guide related modules
- Administrators retained exclusive access to verification and management operations

The token-based authentication mechanism successfully maintained secure user sessions throughout system usage. Unauthorized access attempts were effectively restricted through role validation procedures.

These results demonstrate that the system provides a secure operational environment while maintaining

5.7 Risk Management Plan and Mitigation Plan Control

A risk management plan is an essential component of the project development process, as it helps identify potential risks that may affect the successful implementation of the system. The purpose of this plan is to analyze possible technical, operational, and security-related challenges that could arise during the project lifecycle. By evaluating the probability and impact of each risk, appropriate mitigation strategies can be established to minimize negative effects on the project.

Identification of Risk	Risk Level	Probability for Occurrence of Risk	Mitigation Plan
Difficulty in implementing multilingual support for Sinhala and Tamil languages	High	High	Limit the initial implementation to English language support and plan future integration of multilingual NLP models and datasets.
AI module generating inaccurate or incomplete responses	High	Medium	Implement structured prompt, validation mechanisms, and clarification prompts to improve AI response accuracy.
Failure of external AI APIs or third-party services	High	Medium	Use error handling mechanisms, fallback responses, and maintain alternative API integration strategies.
Data privacy breaches or unauthorized access to user information	High	Medium	Apply token based authentication, encrypted communication, secure database storage, and privacy-locking mechanisms.

Fraudulent tour guide registrations using fake documents	High	Medium	Implement multi-step manual verification processes including ID validation, tourist board license checks, and police clearance verification.
Payment transaction failures during booking confirmation	Medium	Medium	Integrate secure and reliable payment gateways such as Stripe and implement payment validation before booking confirmation.
System downtime or backend server failure	High	Low	Use scalable cloud infrastructure, regular backups, monitoring systems, and redundancy mechanisms.
Poor internet connectivity affecting real-time system interaction	Medium	Medium	Optimize frontend performance, reduce unnecessary API calls, and implement efficient loading and caching mechanisms.
Incompatibility between frontend, backend, and external APIs	Medium	Medium	Conduct integration testing regularly and follow standardized API communication protocols.

Table 3 - Risk Management Plan

Discussion

The results obtained from the evaluation demonstrate that the proposed Safari platform successfully satisfies the major functional requirements identified during the requirement gathering and system design phases. The developed system not only provides standard booking functionalities but also addresses several weaknesses commonly observed in existing tourism platforms, particularly in the areas of personalization, trust, privacy, and safety. Through the integration of artificial intelligence and secure workflow mechanisms, the platform introduces a more intelligent and user-centric approach to digital tourism services.

One of the most significant outcomes of the project is the successful implementation of AI-driven conversational interaction. Traditional tourism booking systems generally depend on static forms where users are required to manually enter large amounts of information through predefined fields. This process can often be time-consuming, complex, and inconvenient, especially for users with limited technical knowledge. In contrast, the proposed system allows users to interact with the platform using natural language conversations. Users can simply describe their travel requirements, including destination, budget, duration, and preferences, in a conversational manner.

The AI module analyzes these inputs and dynamically extracts the required information to create structured tour requests. This conversational approach improves usability by reducing the effort required from users and making the interaction process more intuitive. Furthermore, the AI system enhances personalization by generating recommendations and itinerary suggestions that align with user preferences. The evaluation results indicate that users were able to complete booking-related tasks more efficiently through AI-assisted interaction compared to traditional manual input methods. These findings confirm the effectiveness of conversational AI in improving user experience within tourism-related digital platforms with itinerary plans.

Another important contribution of the system is the implementation of verification mechanisms and privacy-aware workflows, which significantly improve trust and security within the platform. Existing tourism platforms often rely heavily on user reviews and ratings as the primary method for establishing trust. However, review-based systems may be vulnerable to manipulation, fake accounts, and fraudulent activities. To address this issue, the proposed platform introduces a multi-step verification mechanism for tour guides.

Under this mechanism, guides are required to submit official documents such as national identity cards, tourist board licenses, and police clearance certificates. Administrators manually review and approve these documents before granting platform access. This process ensures that only verified and legitimate guides can participate in booking activities. As a result, users can interact with greater confidence, knowing that the platform actively validates service providers.

In addition to verification, the platform incorporates a privacy-locking mechanism that protects sensitive user information during the booking process. Personal details such as contact information remain hidden until a booking is officially confirmed. This conditional data-sharing approach minimizes risks related to unauthorized communication, data misuse, and privacy breaches. Together, these features establish a safer and more trustworthy digital environment compared to traditional tourism systems.

The successful integration of booking workflows, emergency handling, and secure payment support further demonstrates the practical feasibility of the proposed system. The booking workflow effectively manages the complete lifecycle of a booking process, including request creation, guide bidding, booking confirmation, and payment validation. Testing confirmed that these workflows functioned consistently without critical failures, indicating strong coordination between frontend interfaces, backend services, and database operations.

The integration of secure payment processing ensures that financial transactions are handled safely and reliably, contributing to user confidence during online payments. Additionally, the emergency alert feature enhances user safety by enabling tourists to send emergency notifications directly to administrators during active bookings. This feature is particularly important in the tourism domain, where traveler safety is a major concern. The inclusion of such safety oriented functionalities demonstrate the system’s capability to operate as a comprehensive tourism assistance platform tailored with itinerary plans to the Sri Lankan context.

Despite the successful implementation of major system functionalities, certain limitations were identified during the evaluation process. The AI module occasionally required additional clarification prompts when processing ambiguous, incomplete, or unstructured user inputs. Although the AI system performed effectively in most scenarios, there were instances where it could not immediately determine user intent with complete accuracy. In such cases, the system prompted users for additional information before proceeding. This limitation highlights the challenges associated with natural language understanding, particularly in conversational AI systems. It also indicates the need for improved contextual understanding and more advanced dialogue management capabilities in future versions of the platform.

Overall, the findings confirm that the proposed Safari platform provides a secure, intelligent, and operationally reliable solution capable of enhancing digital tourism services. The integration of AI-driven interaction, structured verification mechanisms, privacy-aware workflows, and secure booking management demonstrate the practical value and technical feasibility of the system. The results also suggest that intelligent tourism platforms can significantly improve user experience, trust, and operational efficiency within the tourism industry, particularly in the Sri Lankan context

6. Future Scope

6.1 Ai Agent Support and Itinerary Plan

One of the primary future improvements of the system is the implementation of full multilingual support, particularly for Sinhala and Tamil languages in addition to English. The current system mainly supports English due to technical limitations associated with NLP for low-resource languages.

Future enhancements could involve the integration of advanced multilingual AI models capable of understanding and processing Sinhala and Tamil inputs more effectively. This would require the development or integration of language-specific datasets, translation pipelines, and fine-tuned conversational models. Supporting multiple languages would improve accessibility and inclusiveness, enabling a wider range of local and international users to interact with the system more naturally.

Furthermore, multilingual voice interaction could be introduced through speech recognition and speech synthesis technologies, allowing users to communicate with the platform using voice commands in their preferred language.

6.2 Advanced AI and Recommendation Capabilities

The AI module can be further enhanced by incorporating more advanced recommendations and contextual understanding capabilities. Currently, the system generates recommendations based on user-provided information such as destination, budget, and travel duration. However, future versions could integrate:

- Real-time weather information
- Seasonal tourism trends
- User behavior analysis
- Historical booking patterns
- Personalized preference learning

By integrating machine learning and predictive analytics, the system could provide more intelligent and adaptive recommendations over time. Future AI enhancements may also include sentiment analysis and emotion-aware interactions to improve conversational quality and user engagement.

Additionally, improvements in contextual memory and dialogue management could reduce the need for clarification prompts and enhance the overall conversational experience.

6.3 AI-Based Image Generation for Itinerary Planning

Future versions of the platform could incorporate AI-generated visual itinerary planning, allowing users to view destination images, attraction previews, and travel route visualizations alongside textual recommendations.

This functionality could be implemented using advanced text-to-image generative AI models capable of creating contextual travel-related visuals based on user preferences and itinerary details. Such visual enhancements would improve user engagement and provide a more interactive travel planning experience.

Furthermore, integrating image generation with itinerary suggestions could assist tourists in making better travel decisions by providing visual representations of destinations, accommodations, and activities.

6.4 Mobile Application Development

Currently, the system is implemented primarily as a web-based platform. Future work could include the development of dedicated mobile applications for Android and iOS devices.

A mobile application would improve accessibility, portability, and user convenience by enabling travelers to access platform services while traveling. Mobile integration could also support additional features such as:

- Push notifications
- GPS-based location services
- Real-time navigation assistance

These features would significantly enhance the practical usability of the system in real-world tourism scenarios.

7. Commercialization

The commercialization strategy for the AI Powered Tour Guide Booking System with Voice and RAG is designed to ensure both sustainability and scalability. The platform will initially launch in Sri Lanka, targeting popular tourist destinations such as Colombo, Kandy, Galle, and Anuradhapura. Once validated locally, the system can be expanded regionally and globally, offering a strong competitive edge in the digital tourism market.

Revenue will primarily be generated through three models:

- Commission Based Revenue:

  A 10% to 15% commission charged on each successful booking between tourists and guides.
- Subscription Plans for Guides:

  Premium accounts offer enhanced visibility, advanced analytics, and featured listings for verified guides.
- Partnerships and Collaborations:

  Strategic partnerships with hotels, airlines, and travel agencies to integrate the system into tourism packages.

Future opportunities include expanding into value added services such as travel insurance, curated experiences, and AI based itinerary planning. The system's scalability, strong verification model, and emphasis on privacy and safety position as a trusted global tourism solution, ensuring long term profitability.

8. Budget

The deliverable of the proposed model is a software based solution, there are no Hardware components connected to the implementation. The biggest source of the cost will be the personnel costs, hardware and software for the computing power of the machine system. However, there will be some other costs expected that will be given in the table below.

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Type	Cost (LKR)
Personnel Costs	70 000
Model Development and Training	50 000
Integration and Deployment	60 000
Project Management and Administration	20 000
Total Project Budget	200 000

Table 2: Cost Management Plan

9. Limitation

Although the proposed Safari AI driven tour guide booking platform successfully achieves its primary objectives, several limitations were encountered during the development and implementation phases. These limitations are mainly associated with multilingual capability and advanced feature integration. Identifying these constraints is important for understanding the scope of the current system and for guiding future enhancements.

9.1 Limited Multilingual Language Support

One of the major limitations of the system is the inability to fully implement multilingual support, specifically for Sinhala and Tamil languages. While the system was initially designed to support three languages Sinhala, Tamil, and English the final implementation was restricted to English language functionality.

From a technical perspective, the integration of Sinhala and Tamil languages presents several challenges due to their classification as low resource languages in the field of Natural Language Processing (NLP). Unlike English, which benefits from extensive pretrained models, large scale datasets, and well established linguistic tools, Sinhala and Tamil lack sufficient resources for effective AI model training and deployment.

Key technical challenges include:

- Difficulty in achieving accurate intent recognition and entity extraction for Sinhala and Tamil inputs
- Limited availability of pre trained language models and annotated datasets
- Complexity in implementing language detection and translation pipelines
- Challenges in maintaining consistent contextual understanding and response generation across multiple languages

These factors significantly increase the complexity of integrating multilingual AI capabilities and require advanced model fine tuning and additional computational resources.

Practical Limitations in addition to technical constraints, several practical challenges were encountered:

- Limited access to domain specific datasets in Sinhala and Tamil
- Difficulty in ensuring accurate translation of tourism related terminology
- Requirement for extensive user testing with native speakers to validate system accuracy
- Increased development time and resource allocation needed for multilingual interface design

Due to these constraints, the implementation of Sinhala and Tamil language support could not be completed within the project timeline. As a result, the system currently supports only English, which may limit accessibility for certain user groups within the Sri Lankan context.

9.2 Limited Image Generation for Trip Itinerary Planning

Another limitation of the system is the absence of dynamic image generation capabilities within the trip itinerary planning module. While the platform effectively generates AI based textual recommendations and structured travel plans, it does not provide visual representations such as images of destinations, activities, or landmarks.

Technical Limitations the implementation of image generation requires the integration of advanced generative AI models, particularly text to image systems. Incorporating such functionality introduces several technical complexities, including:

- High computational requirements for real time image generation
- Need for multi modal AI integration, combining text based and image based models
- Ensuring contextual relevance and accuracy of generated images
- Managing data storage, processing, and delivery of image content

These challenges significantly increase system complexity and require specialized infrastructure and optimization techniques from a practical standpoint, the following constraints affected the implementation:

- Increased infrastructure and operational costs associated with image processing
- Potential issues related to content validation and quality assurance
- Risk of increased system latency, affecting user experience during itinerary generation
- Limited development time and resources to design, integrate, and test image generation features

Due to these factors, the system currently focuses on text based itinerary planning, which ensures efficiency and reliability but lacks visual enhancement.

10. Summary

Tourism is a vital driver of Sri Lanka's economy, yet tourists often face challenges in finding reliable, safe, and verified tour guides. Existing platforms such as TripAdvisor and Viator offer convenience but lack multi level verification, robust fraud detection, privacy aware workflows, and intelligent personalization. These shortcomings reduce trust, create risks, and negatively impact on the tourist experience as well as the country's global reputation.

This project proposes an AI Powered Tour Guide Booking System with Voice and Retrieval Augmented Generation (RAG) to address these gaps. The system will provide multi document verification (ID, licenses, police checks, social media validation), ensuring guide authenticity. A privacy protected workflow will lock tourists' personal details until bookings are confirmed, enhancing security.

A core innovation is the AI driven matching and recommendation engine powered by RAG, which delivers personalized, multilingual responses and matches tourists to suitable guides based on preferences, history, and real time conditions. Safety features such as fraud detection and location tracking further strengthen the platform.

In summary, this project delivers a secure, intelligent, and user friendly tourism platform that builds trust, enhances personalization, and improves safety. It has the potential to transform Sri Lanka's tourism industry and set new standards for digital tourism worldwide.

The AI module demonstrated approximately 80% accuracy in extracting relevant user inputs and guiding interactions. In most cases, the AI effectively identified key parameters such as destination, budget, and travel preferences. For instances where inputs were incomplete or ambiguous, the system employed clarification prompts to refine the data. This highlights the effectiveness of the conversational interaction model, although it also indicates the need for further improvements in handling complex or unclear user queries.

From a system performance perspective, the backend achieved a 97% API success rate, indicating high reliability in processing requests and maintaining stable communication between system components. The use of a layered architecture and efficient session management contributed to consistent performance, even during multi step interactions. Error handling mechanisms ensured that minor issues did not disrupt overall workflows, thereby enhancing system robustness.

Security and privacy features played a critical role in strengthening user trust. The implementation of a multi step verification process for tour guides ensured authenticity and reduced the risk of fraudulent activities. Additionally, the privacy locking mechanism effectively protected sensitive user information by restricting access until booking confirmation. The integrated emergency alert system further enhanced safety by enabling real time communication with administrators during critical situations.

Overall, the results demonstrate that the proposed system provides a more intelligent, secure, and user centric alternative to traditional tourism platforms. The integration of AI driven interactions significantly improves user experience by reducing manual effort and enabling personal recommendations. At the same time, the incorporation of verification and privacy mechanisms addresses key trust and safety concerns.

However, the evaluation also highlights certain limitations. The AI module, while effective, can be further improved to better handle ambiguous inputs and complex conversational scenarios. Additionally, the testing was conducted in a controlled environment with a limited number of scenarios. Expanding the evaluation to real world conditions with a larger user base would provide more comprehensive insights into system performance.

In conclusion, the Safari platform demonstrates strong technical feasibility and practical value, successfully addressing the key challenges identified in existing tourism systems. The findings confirm that integrating artificial intelligence with secure and well structured system design can significantly enhance efficiency, reliability, and user experience of digital tourism services.

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11. References

[1] S. Fernando, J. S. Bandara , and C. Smith, "Tourism in Sri Lanka The Routledge Handbook of Tourism in Asia," 2017,
Available: https://www.academia.edu/download/51847037/Tourism_in_Sri_Lanka__The_Routledge_Handbook_of_Tourism_in_Asia.pdf

[2] M. Zsarnoczky, "HOW DOES ARTIFICIAL INTELLIGENCE AFFECT THE TOURISM INDUSTRY?," Journal of Management, vol. 31, no. 2, 2017.

[3] P. M. B. Jayathilake, "TOURISM AND ECONOMIC GROWTH IN SRI LANKA: EVIDENCE FROM COINTEGRATION AND CAUSALITY ANALYSIS," International Journal of Business, Economics and Law, vol. 2, no. 2, Jun. 2013.

[4] S. Fernando, J. S. Bandara, and C. Smith, "Tourism in Sri Lanka," 2017.

[5] W. Samarathunga, "Post COVID19 Challenges and Way Forward for Sri Lanka Tourism," Apr. 2020, Accessed: Feb. 14, 2026. [Online]. Available: https://ssrn.com/abstract=3581509

[6] L. M. Munasinghe, T. Gunawardhana, and R. G. Ariyawansa, "Sri Lankan Travel & Tourism Industry: Recent Trends and Future Outlook towards Real Estate Development," presented at the 2018 International Conference On Business Innovation (ICOBI), Aug. 2018.

[7] J. A. R. C. Sandaruwani and W. K. A. C. Gnanapala, "THE ROLE OF TOURIST GUIDES AND THEIR IMPACTS ON SUSTAINABLE TOURISM DEVELOPMENT: A CRITIQUE ON SRI LANKA," Tourism, Leisure and Global Change, vol. 3, 2016.

[8] A. N. Kazak, P. V. Chetyrbok , and N. N. Oleinikov, “Artificial intelligence in the tourism sphere,” IOP Conference Series: Earth and Environment, 2020, Available: https://iopscience.iop.org/article/10.1088/1755 1315/421/4/042020

[9] R. Filieri, E. D’Amico, A. Destefanis, E. Paolucci, and E. Raguseo, “Artificial intelligence (AI) for tourism: an European based study on successful AI tourism start ups,” Oct. 2021, Available: https://www.emerald.com/ijchm/article abstract/33/11/4099/126338/Artificial intelligence AI for tourism an European?redirectedFrom=fulltext

[10] A. Ferhataj and F. Memaj, “CHALLENGES AND OPPORTUNITIES OF AI IMPLEMENTATION IN TOURISM: AN ETHICAL AND TECHNOLOGICAL PERSPECTIVE,” 2024, doi: https://doi.org/10.52320/svv.v1iIX.357.

[11] A. Williams and V. Baláž, “TOURISM RISK AND UNCERTAINTY: THEORETICAL REFLECTIONS,” Journal of Travel Research, 2015, Available: https://journals.sagepub.com/doi/abs/10.1177/0047287514523334

[12] A. Lepp and H. Gibson, “TOURIST ROLES, PERCEIVED RISK AND INTERNATIONAL TOURISM,” Annals of Tourism Research, vol. 30, 2003, doi: https://doi.org/10.1016/S0160 7383(03)00024 0.

12. Appendices

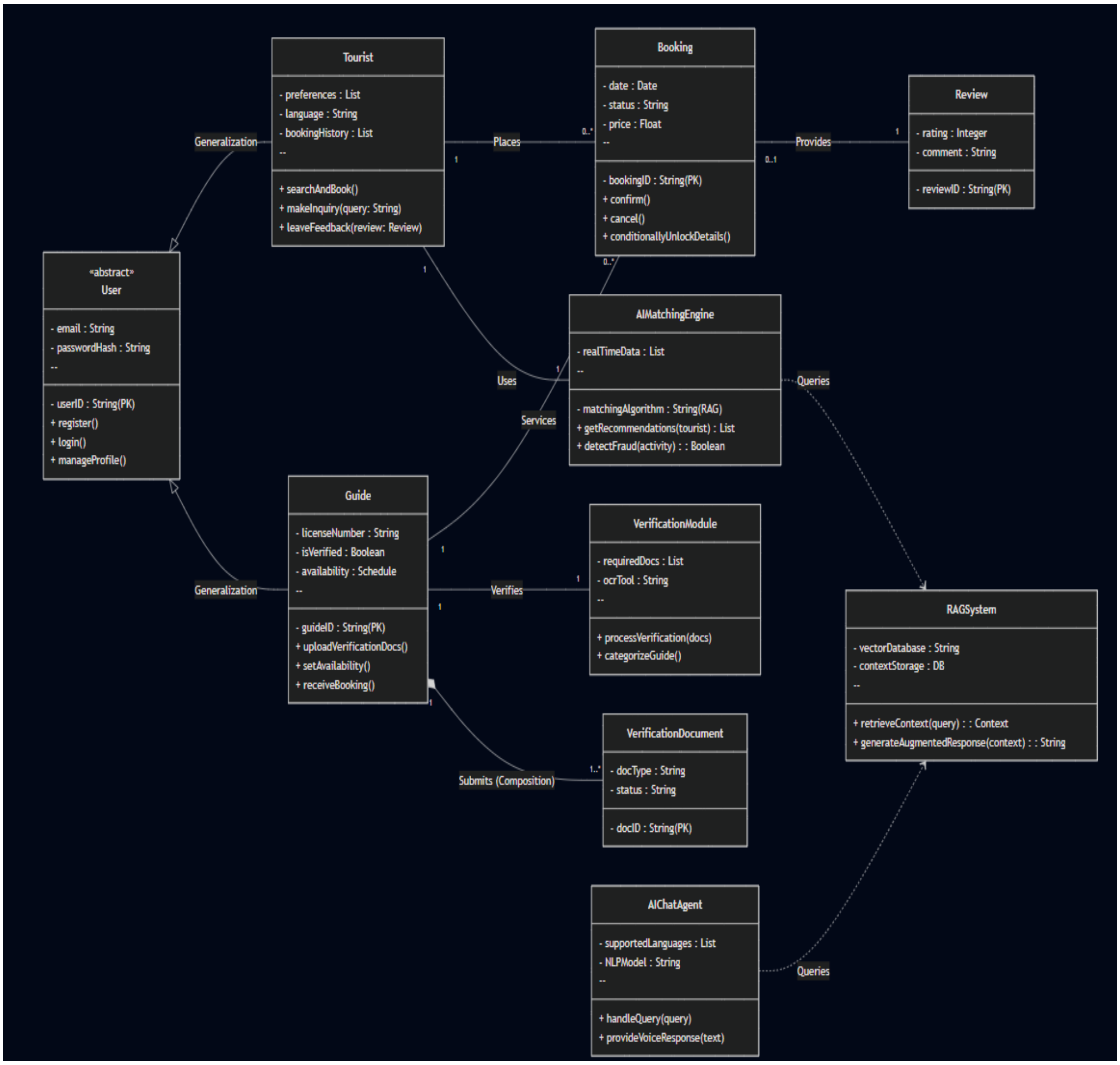

Figure 24 : Class Diagram

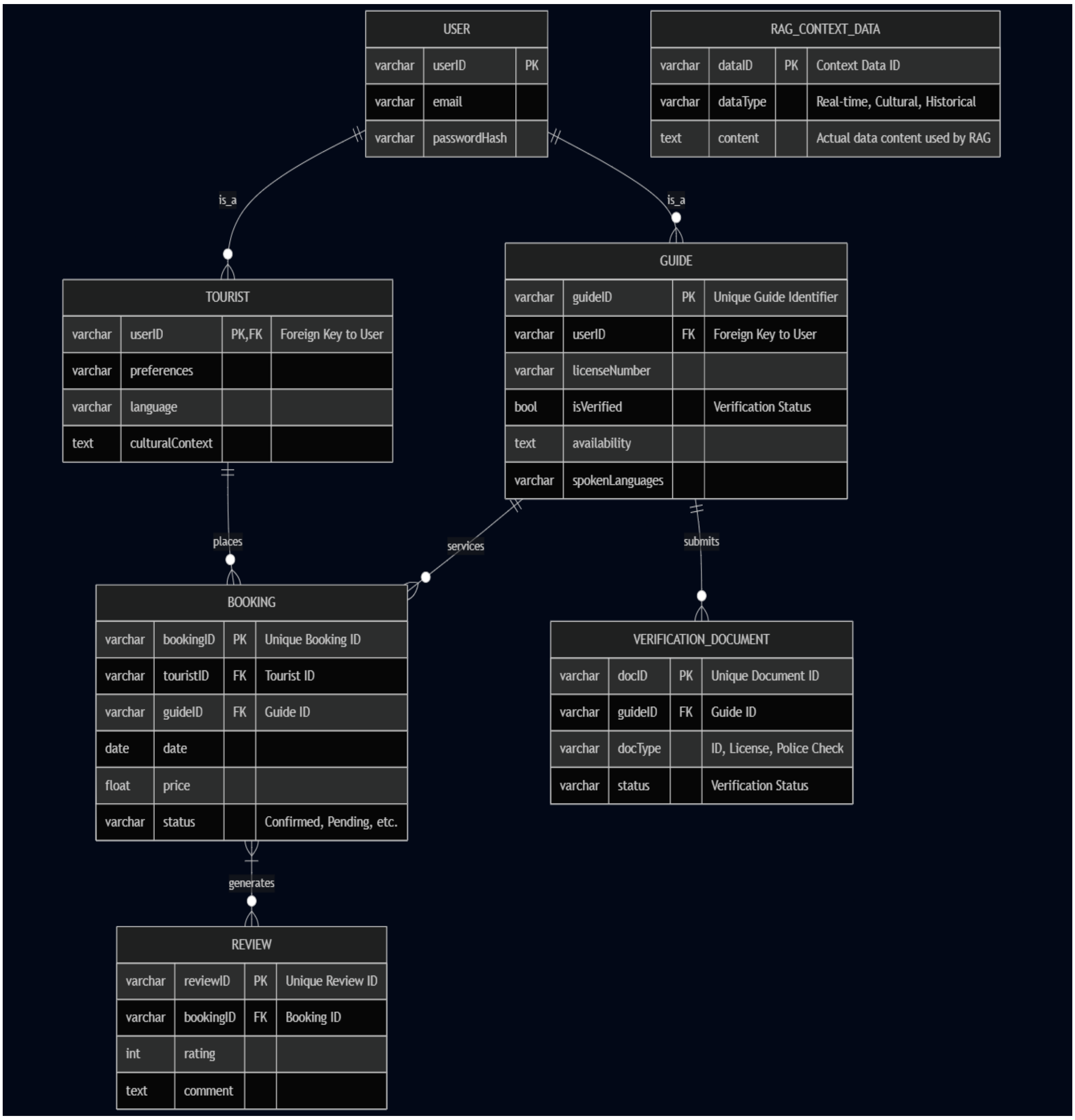

Figure 25 : Relationship Diagram